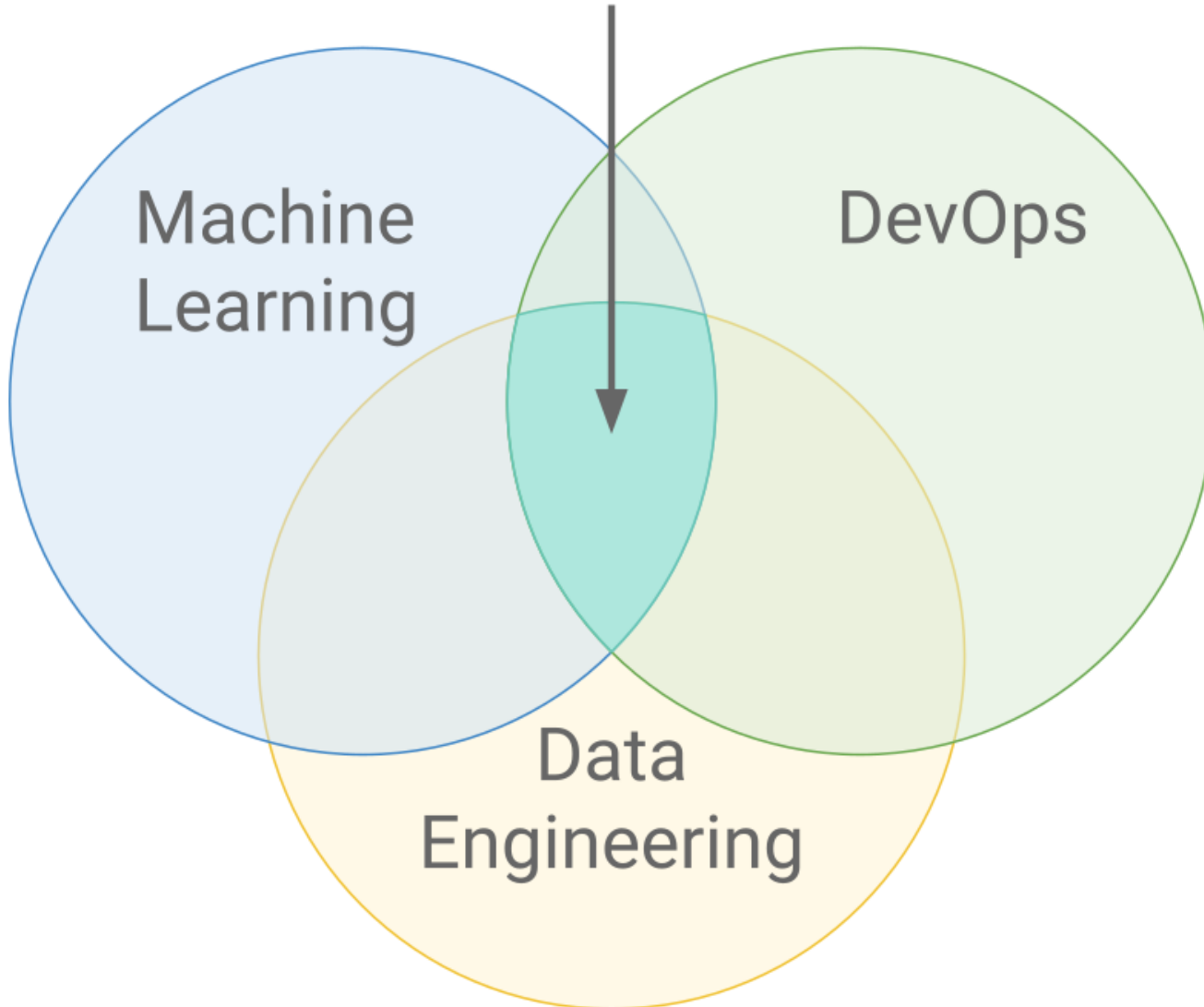


Into the World of MLOps or Machine Learning Operations

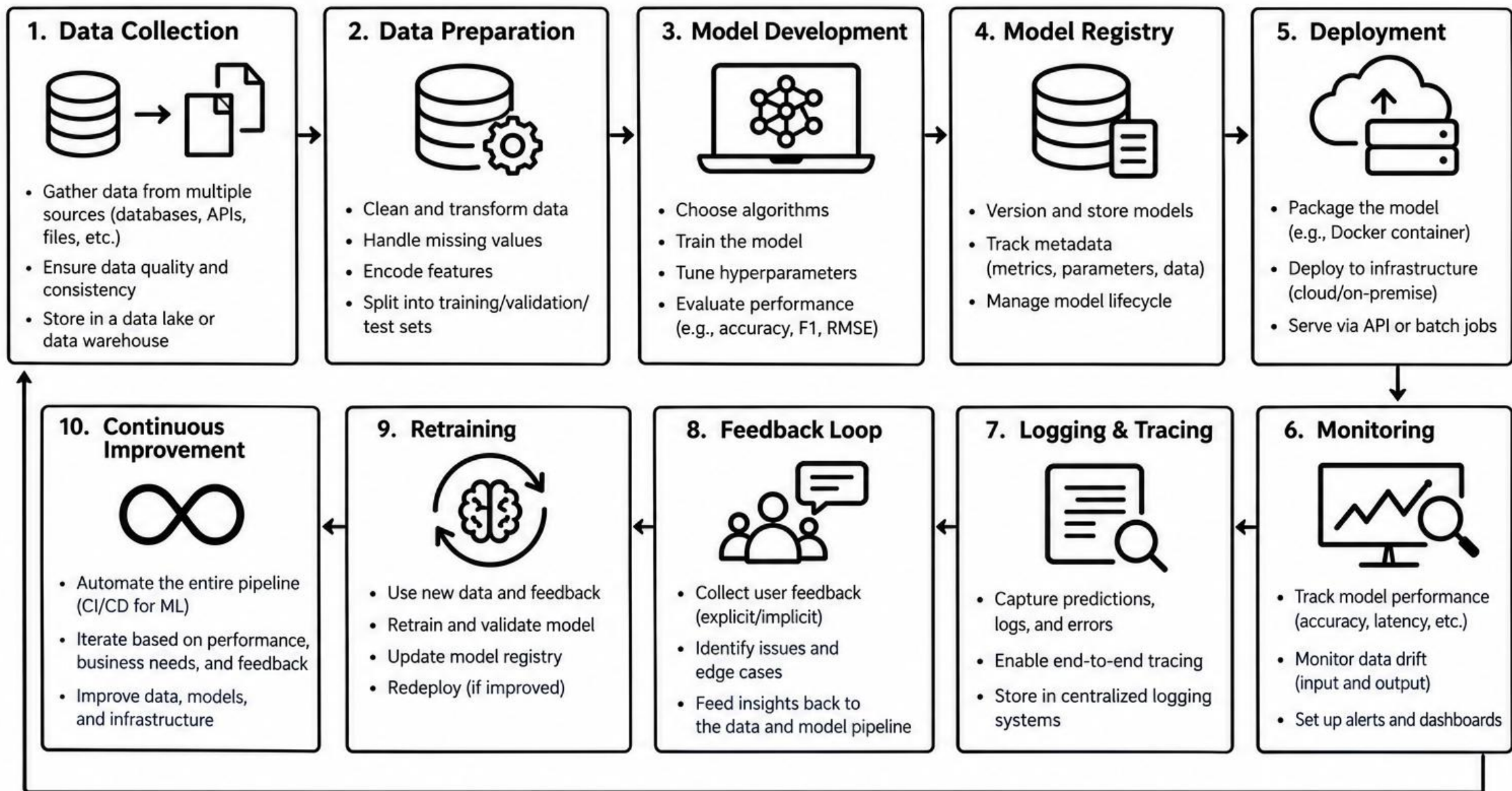
Mohammad Fozouni (Ph.D.)

MLSECOPS IN ACTION COURSE

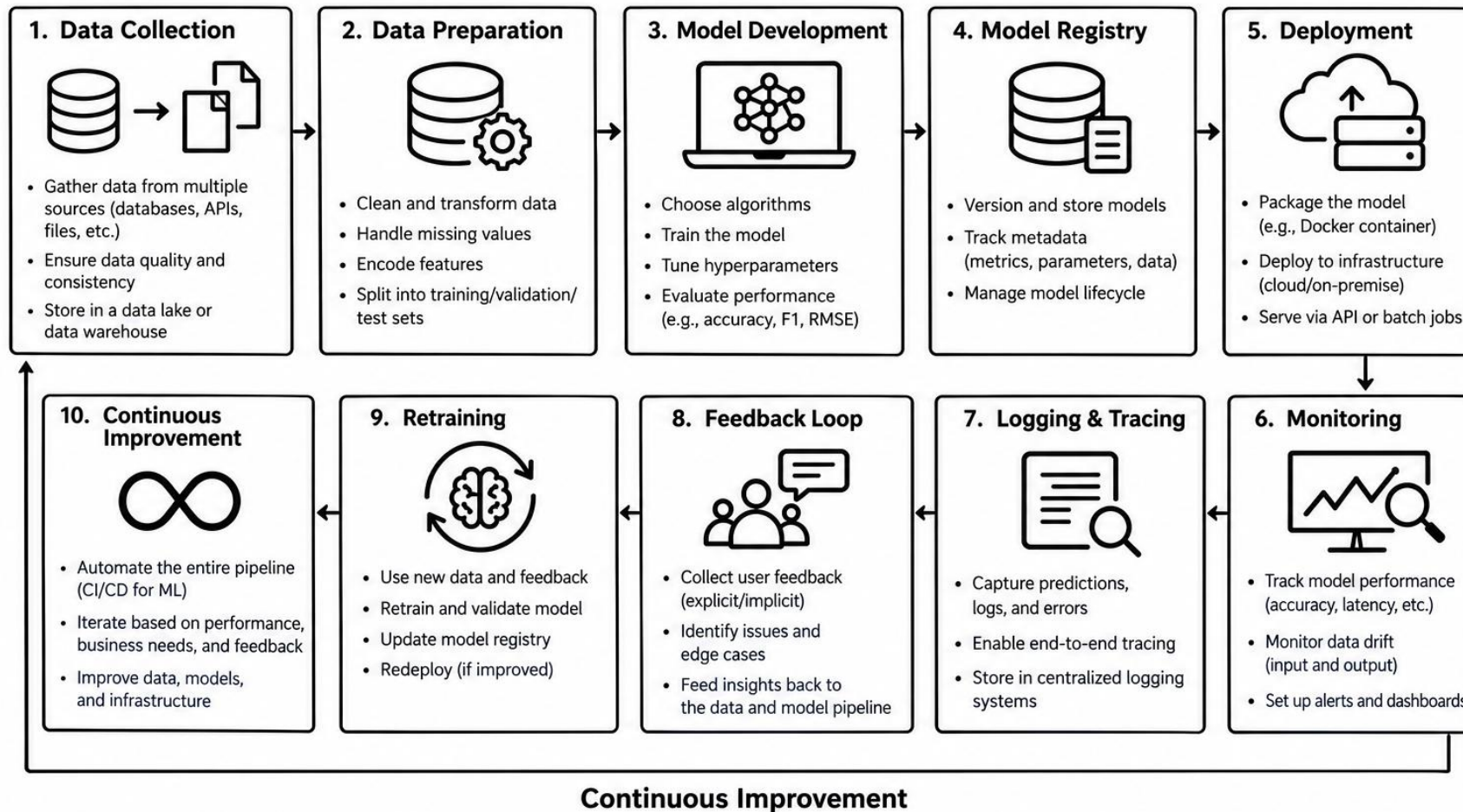
MLOps



**MLOps Engineer could
handle the whole ML
lifecycle.**

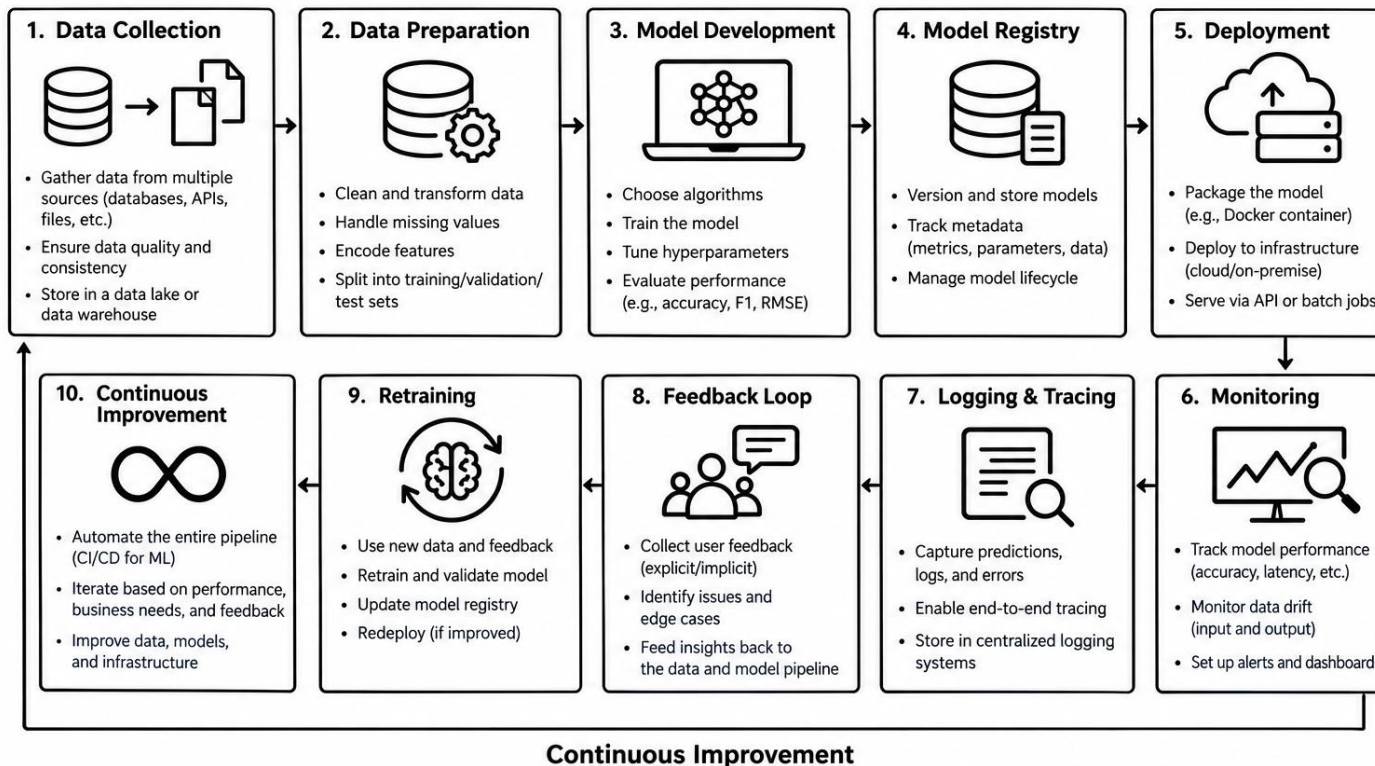


Continuous Improvement



BUTTTTTT, this is not what we should setup from the first day

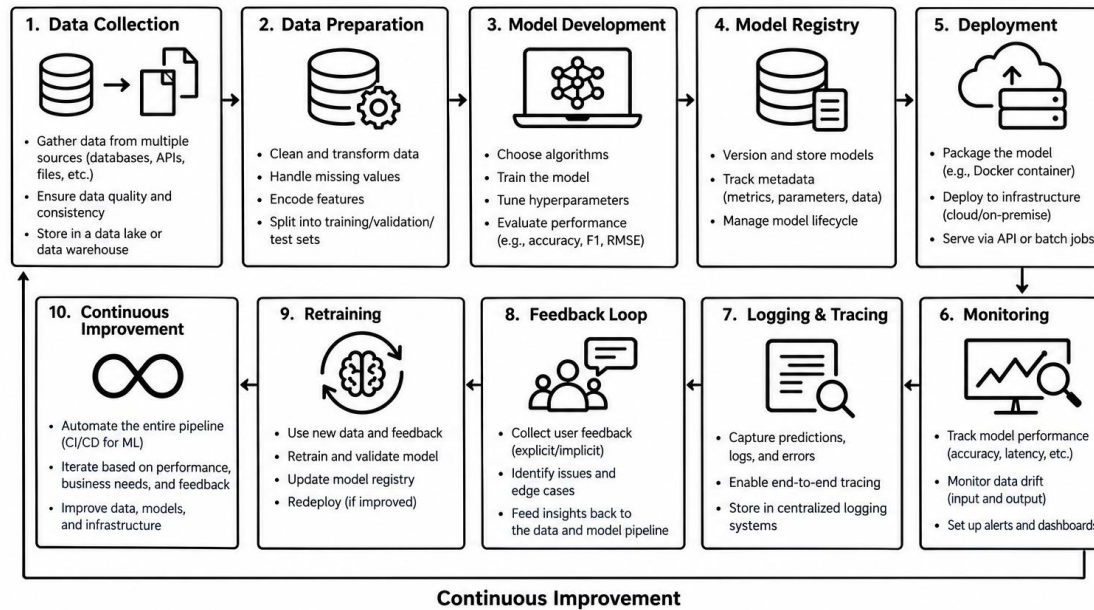
و این چیزی هست که باید پیامـ وزیم!



BUTTTTTT, this is not what we should setup from the first day

و این چیزی هست که باید پیام وزیم!

امال آپس سه سطح داره که باید طبق اون بریم جلو

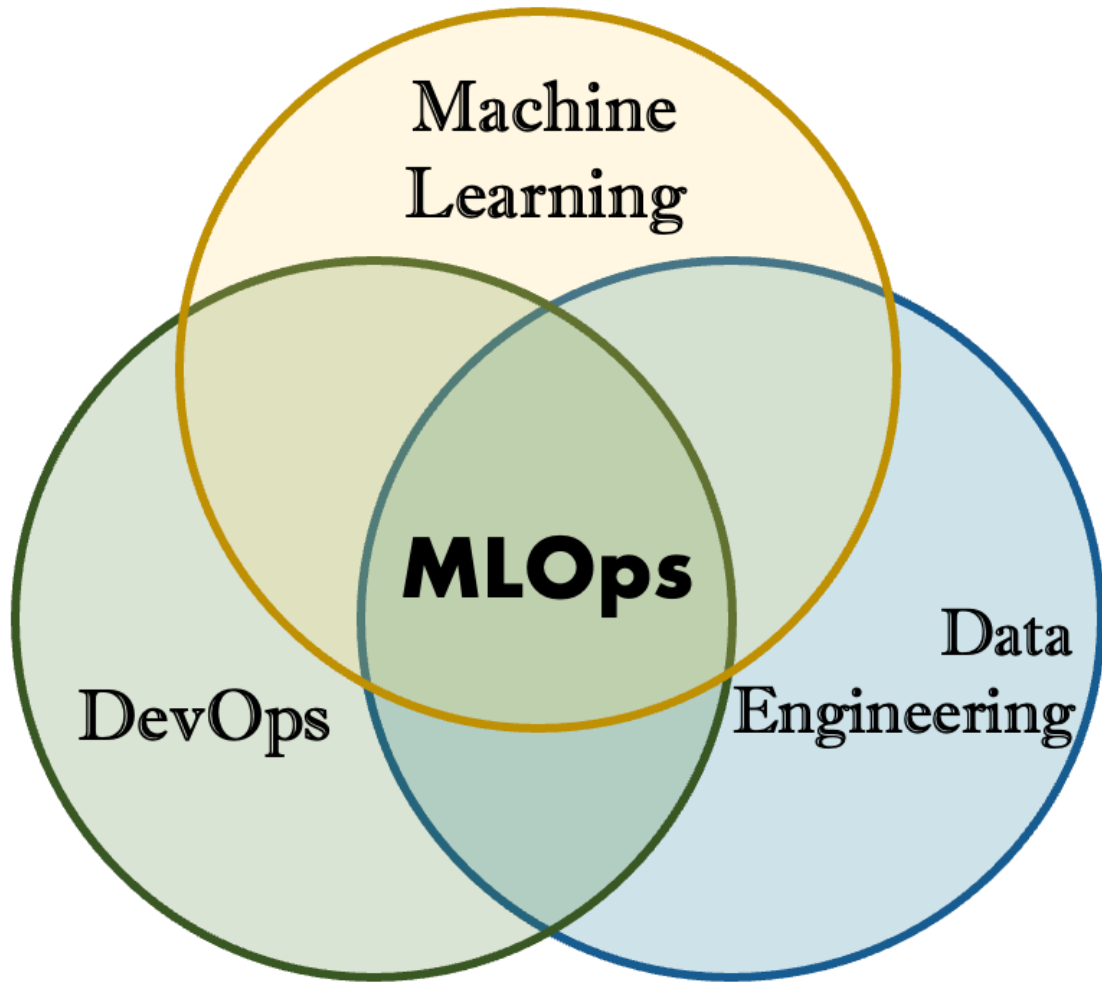


BUTTTTTT, this is not what we should setup from the first day

و این چیزی هست که باید پیام—وزیم!

امال آپس سه سطح داره که باید طبق اون بریم جلو

وگرنه پروژه با شکست مواجه خواهد شد 😞



Which one is more **stressful** in production and why?

1– Data Engineering

2– Machine Learning

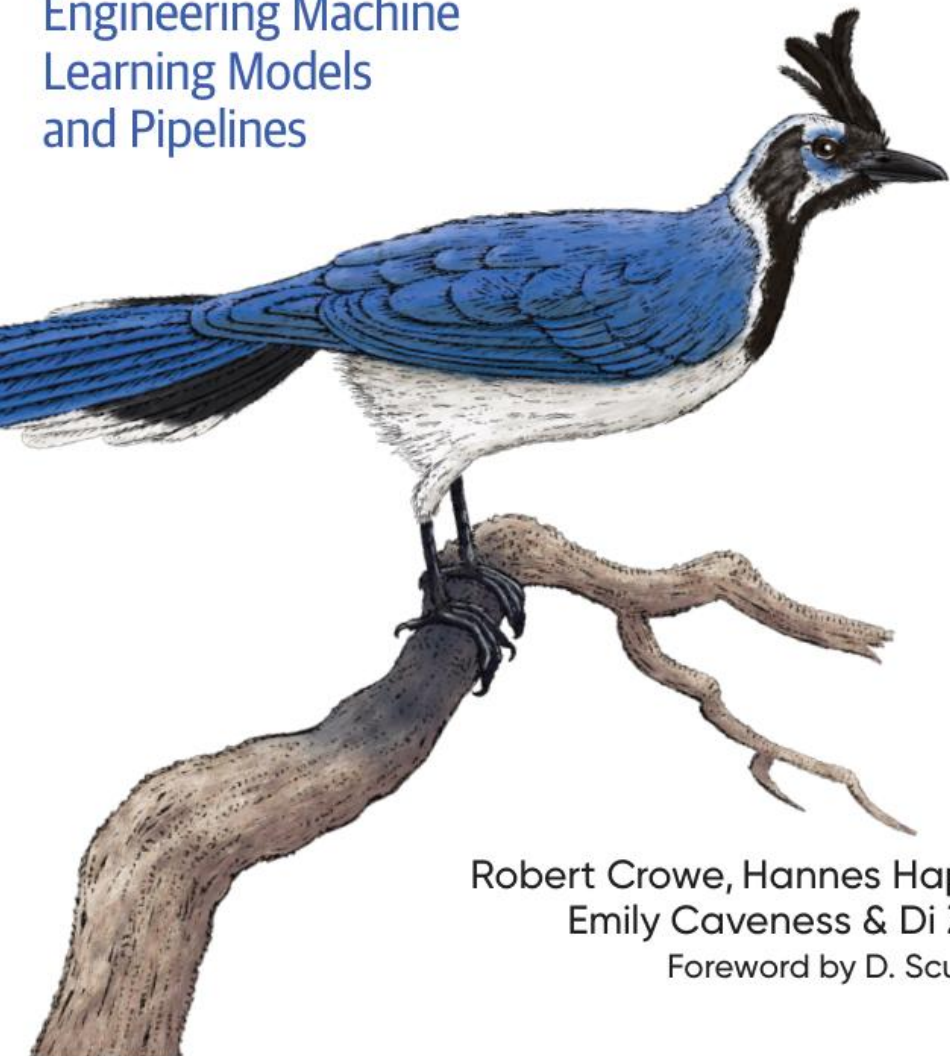
3– DevOps

نکاتی در خصوص منبع علمی این جلسه



Machine Learning Production Systems

Engineering Machine
Learning Models
and Pipelines

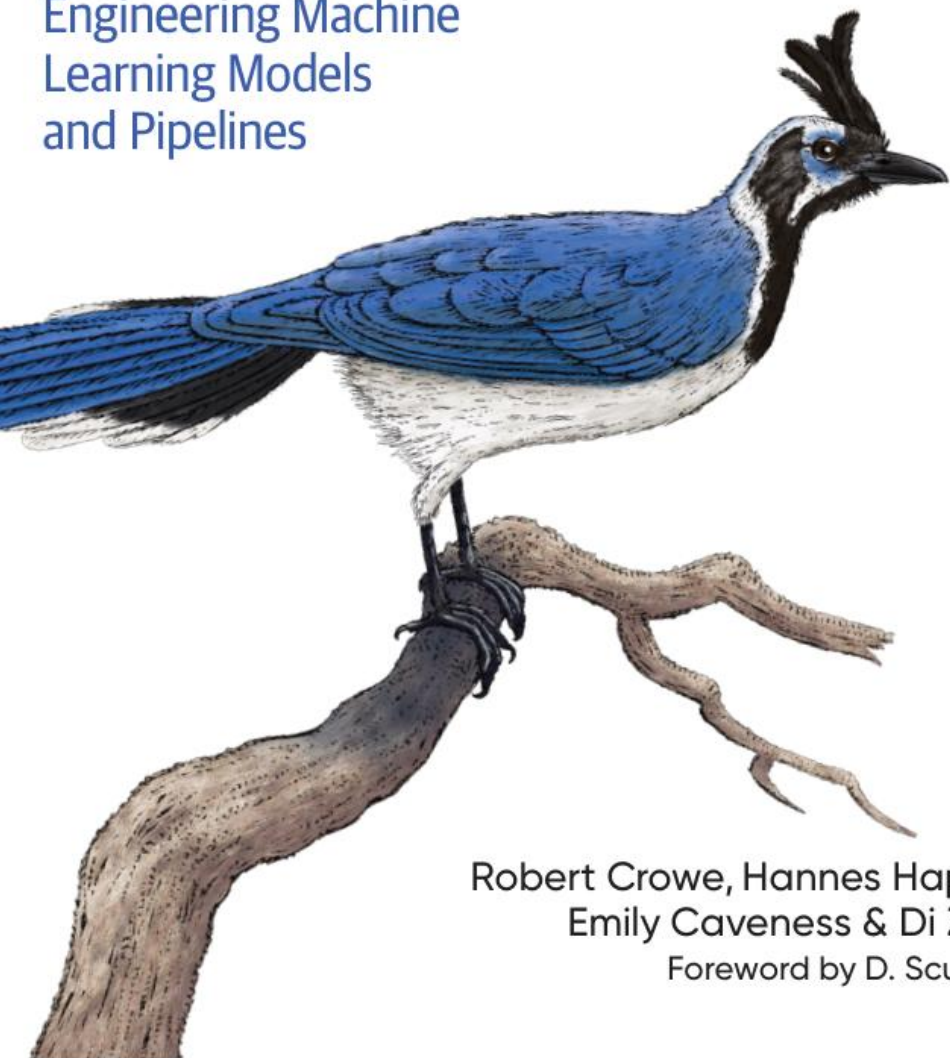


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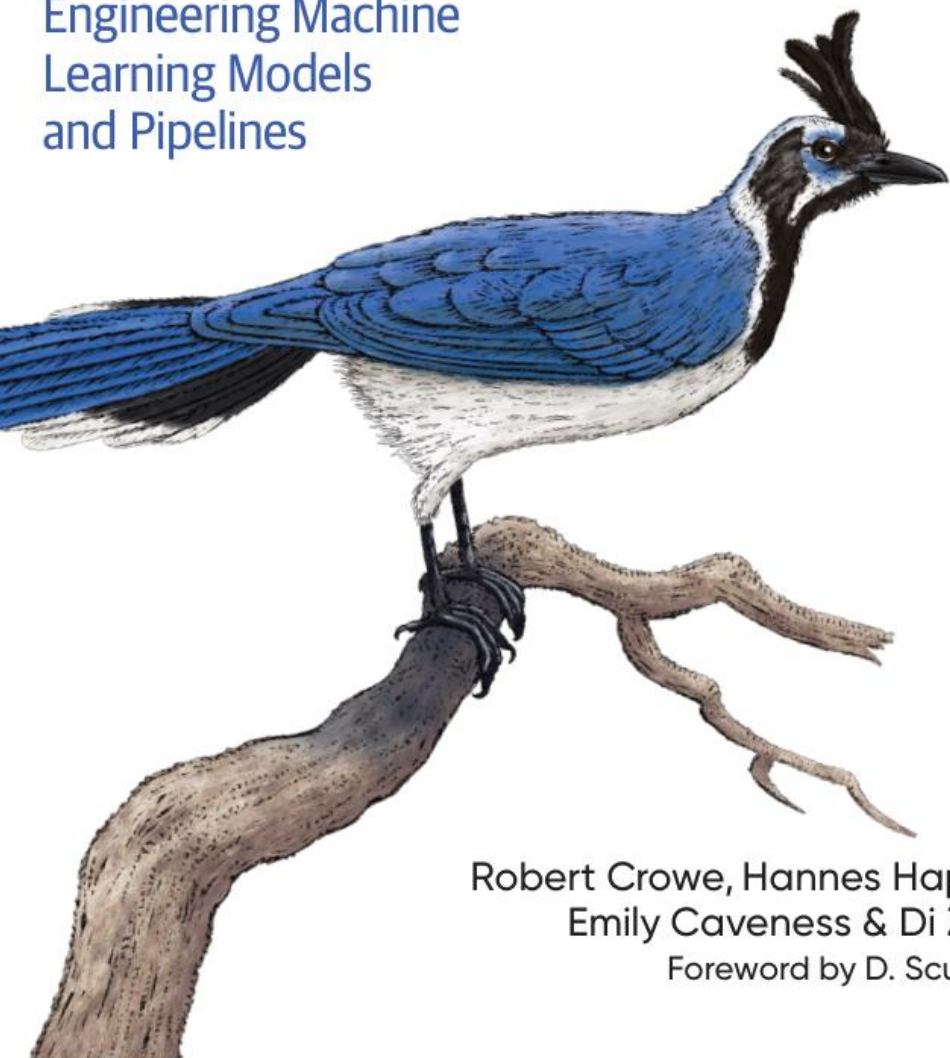


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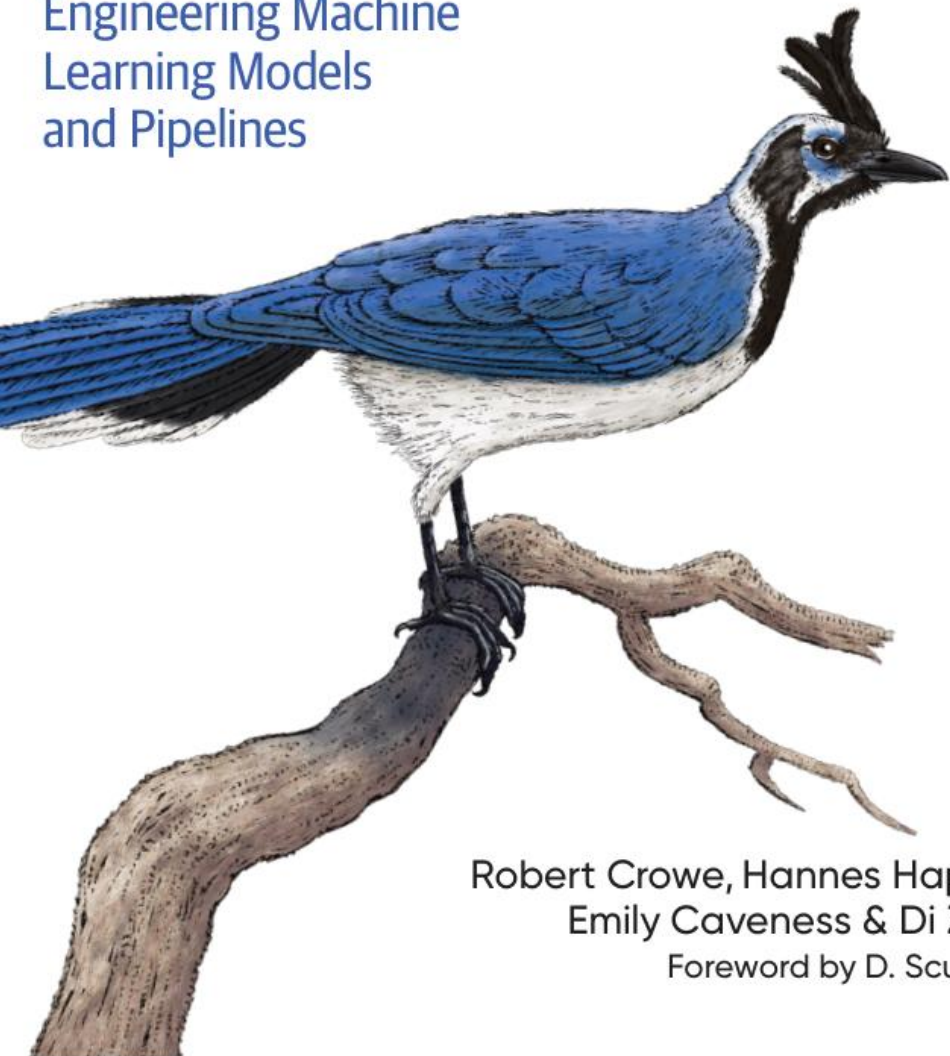


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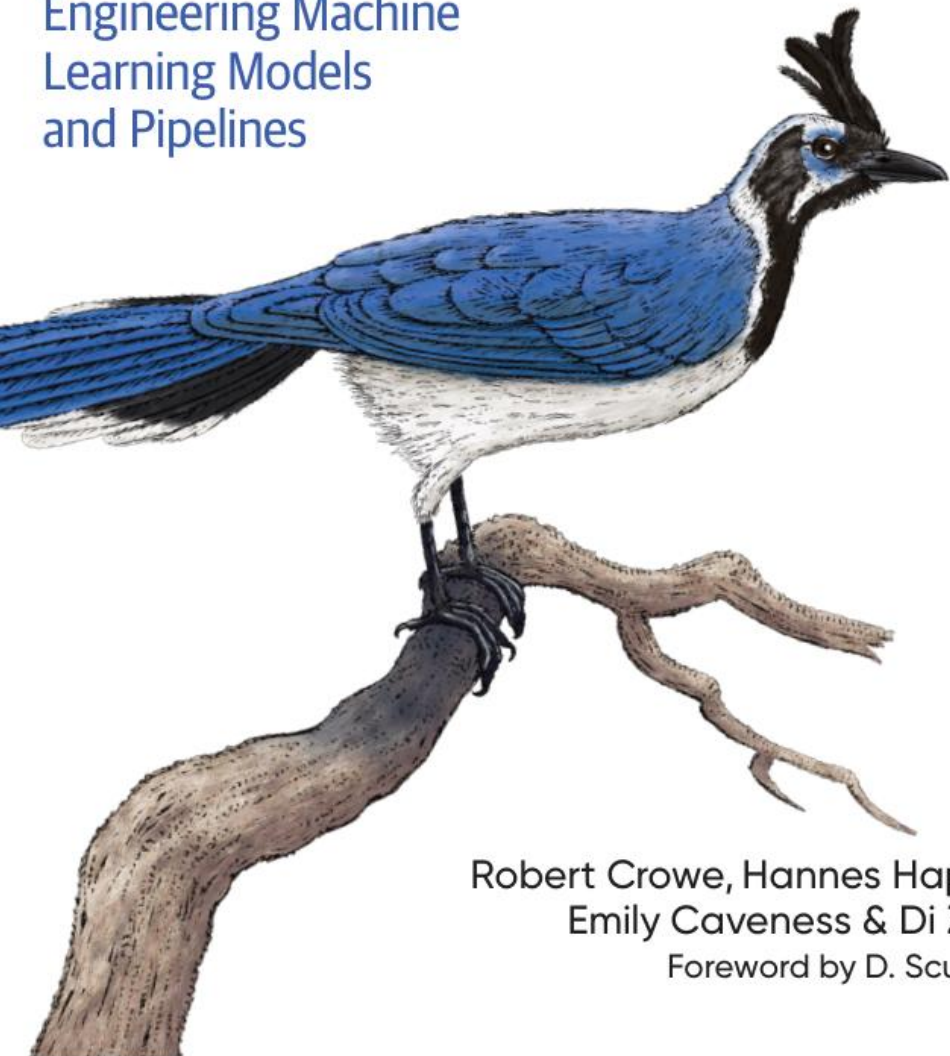


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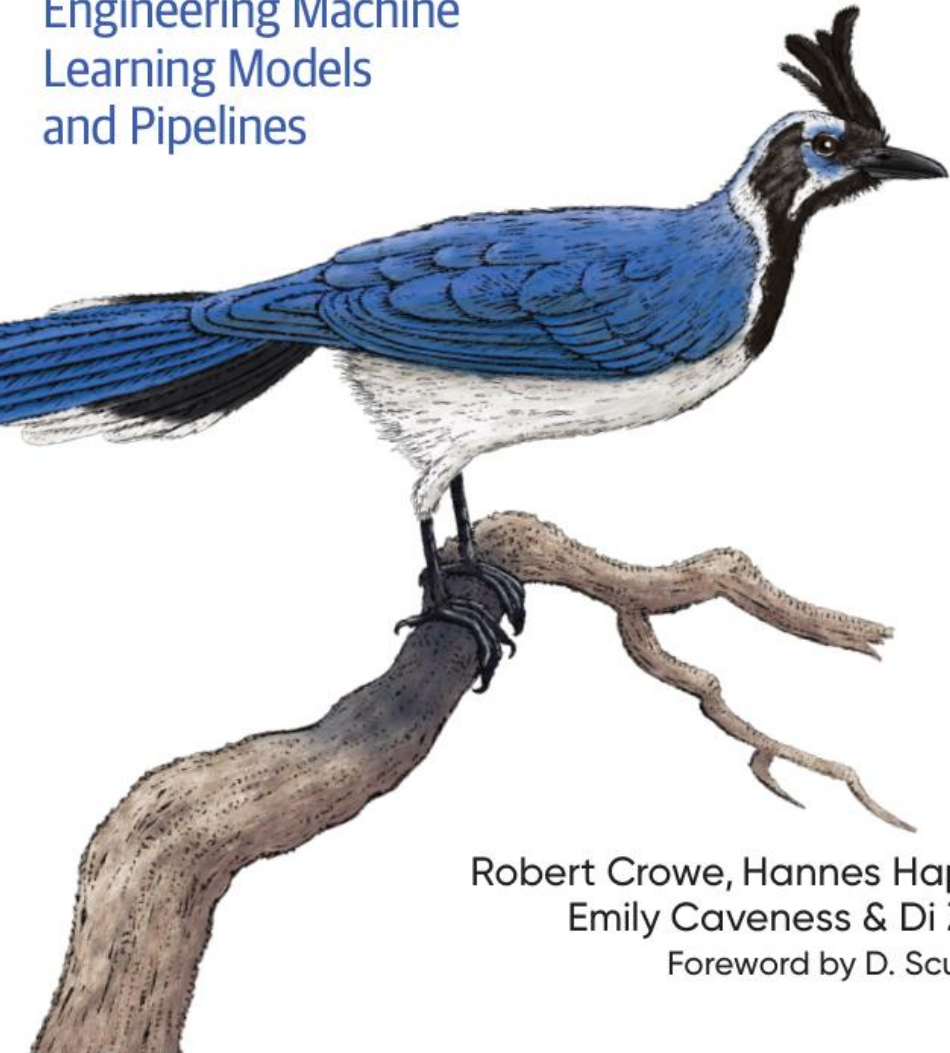


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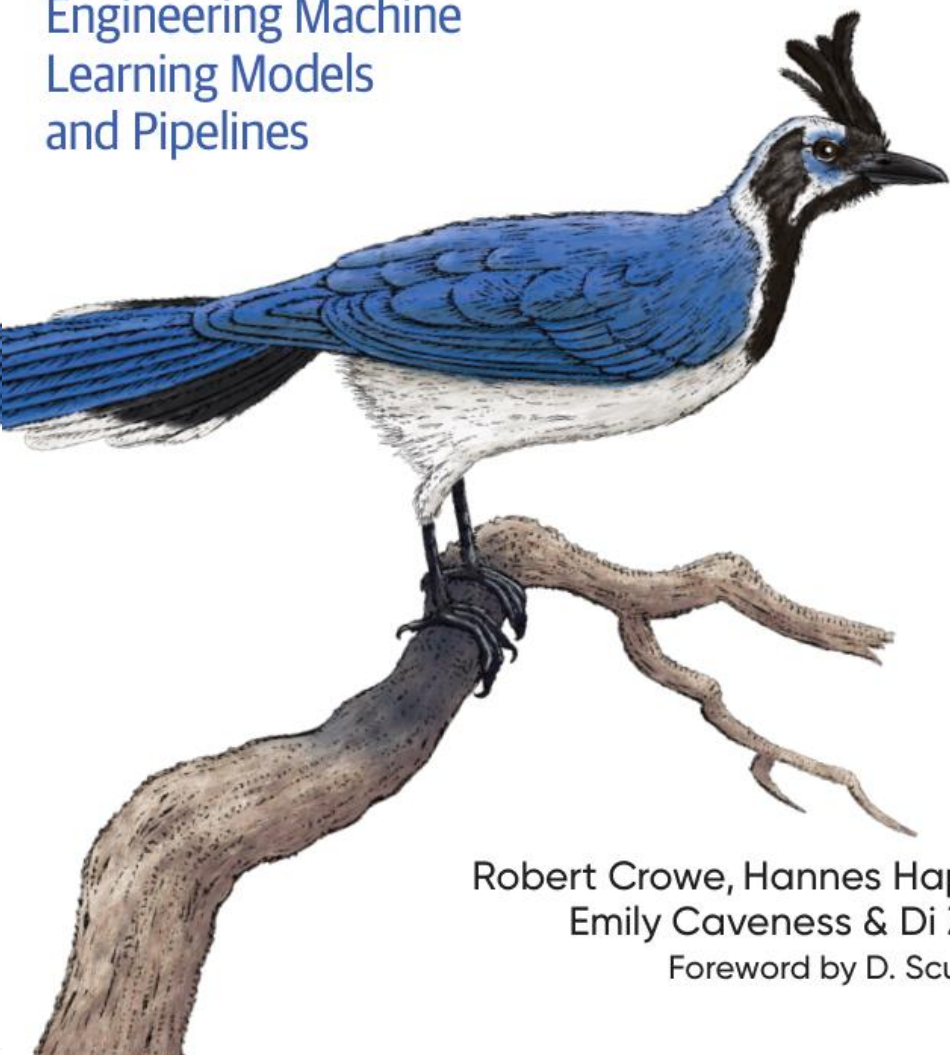


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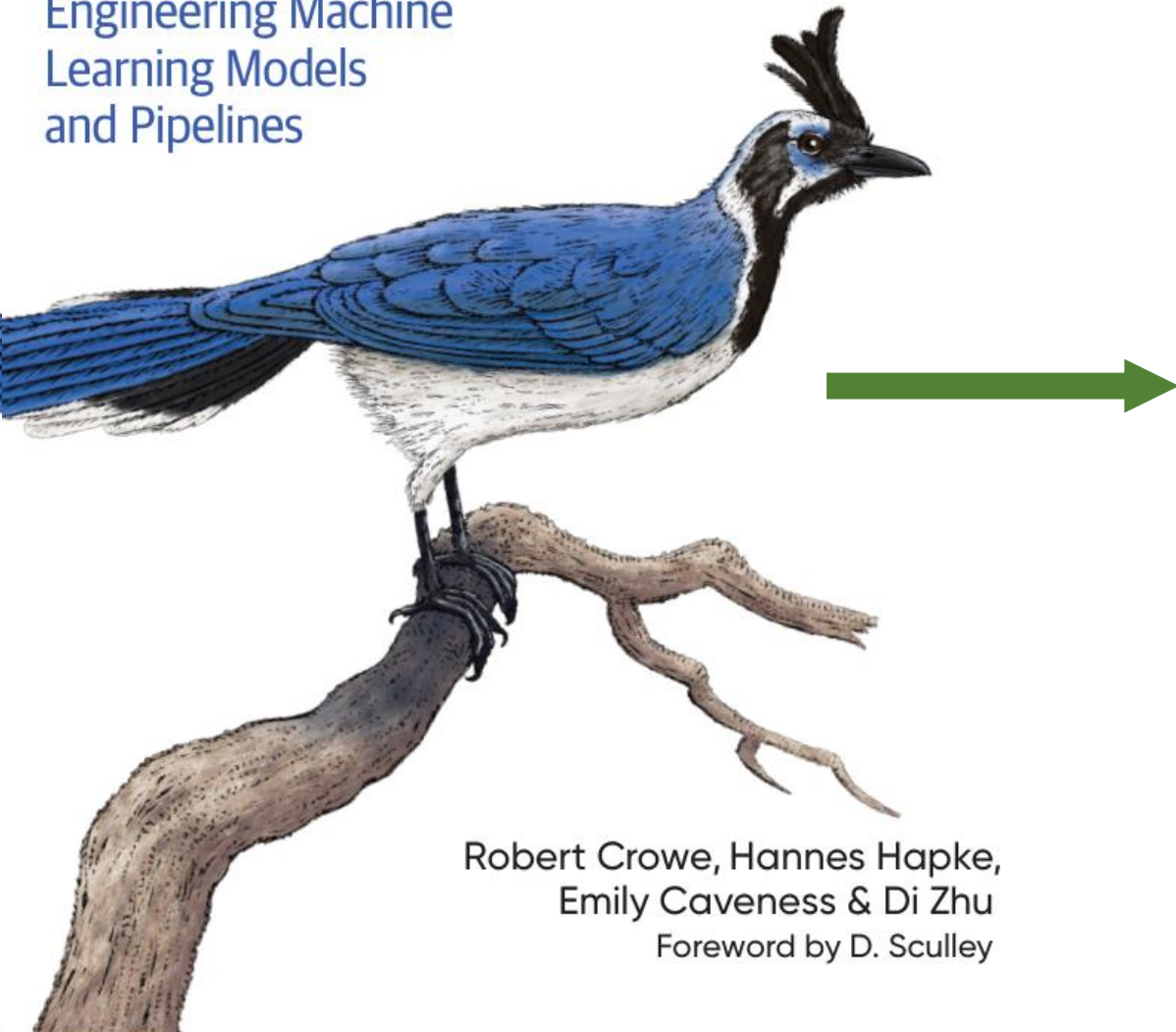


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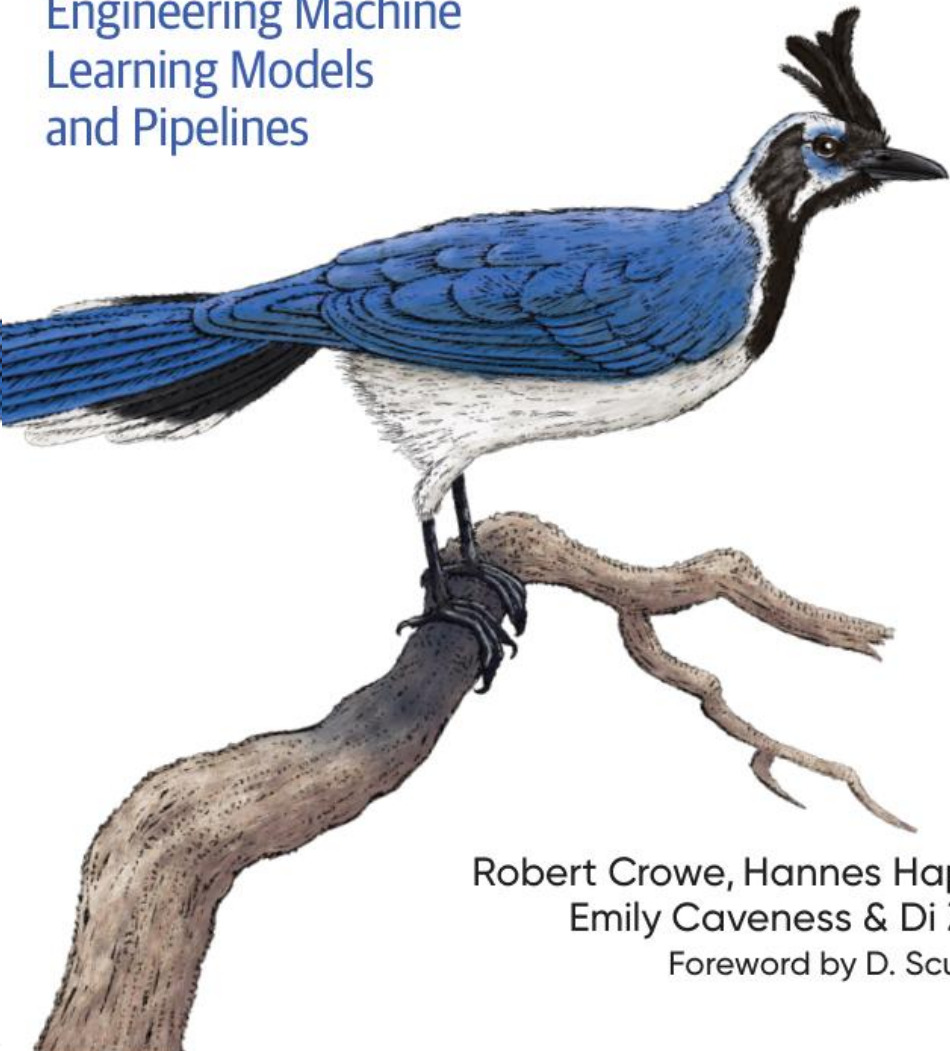
We will talk about these
chapters

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O'REILLY®

Machine Learning Production Systems

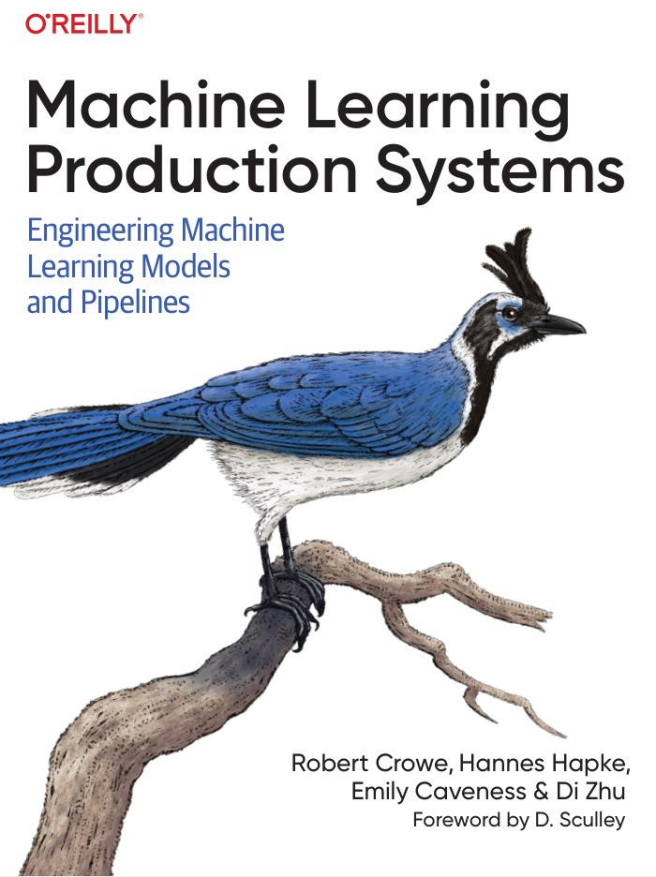
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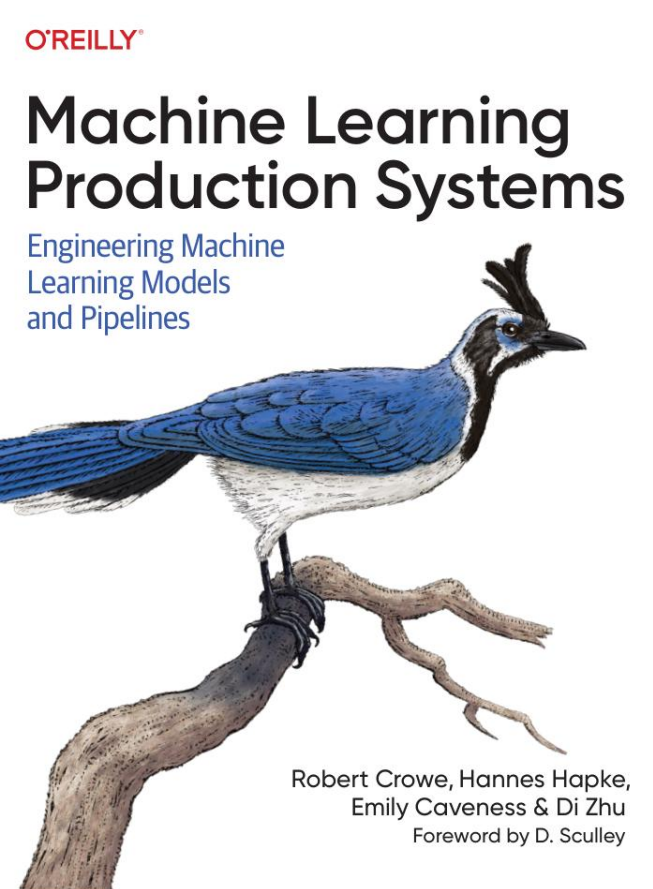
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برخی از ابزارهای
معرفی شده در
کتاب

برخی از ابزارهای معرفی شده در کتاب



Component	Purpose	Key MLOps Use Case
TensorFlow Extended (TFX)	End-to-end ML pipeline framework	Complete MLOps pipeline orchestration
TensorFlow Serving	Model deployment and inference	Production model serving with versioning
TensorFlow Data Validation (TFDV)	Data analysis and validation	Data quality monitoring and drift detection
TensorFlow Transform (TFT)	Data preprocessing at scale	Consistent feature engineering pipeline
TensorFlow Model Analysis (TFMA)	Model evaluation and validation	Performance monitoring and model comparison
TensorFlow Metadata (TFMD)	Metadata management	Pipeline tracking and reproducibility
TensorFlow Hub	Model repository	Reusable models and transfer learning
TensorFlow Lite	Mobile/edge deployment	Model optimization for resource constraints
TensorBoard	Visualization and monitoring	Experiment tracking and performance analysis
Kubeflow	Kubernetes-based ML workflows	Cloud-native pipeline orchestration



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یک نکته احتمالن منفی در خصوص کتاب

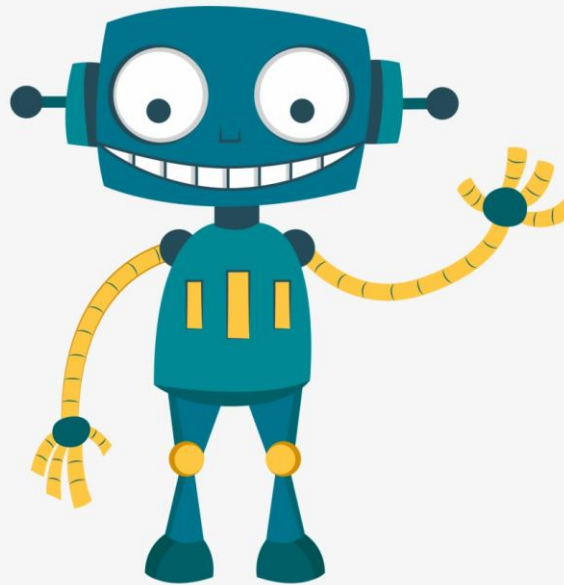
The exclusive focus on **TensorFlow** may be a limitation for some readers. That said, the authors draw from real-world experience, making each insight in this book genuinely valuable.

خلاصه برخی از فصل‌های کتاب



Chapter 11

Introduction to Model Serving



Prediction $\cong$ Result $\cong$ Inference

Two basic types of **model training**

1. **Offline training:**

The model is trained on a set of already collected data. After deploying to the production environment, the model remains frozen until it is retrained with new data.

2. **Online training:**

The model is regularly being updated as new data arrives.

Two basic types of **model prediction**

1. **Batch predictions:**

The deployed model makes a set of predictions based on a batch input data containing multiple examples.

2. **Real-time predictions (aka on-demand predictions):**

Predictions are generated in real time using the input data that is available at the time of the request. This is often used in cases where users or systems are blocked, waiting for the model results.

Three concerns in deploying ML Models

1. **Latency:** Latency is the time delay between sending a request to a model and receiving a result.
2. **Throughput:** Throughput is the number of successful requests served per unit of time, often measured as queries per second (QPS). In some applications, throughput is much more important than latency.
3. **Cost:** We should always try to minimize the cost associated with each inference, to the extent that the inference still meets the business needs. For a trained model to be viable for a business, the cost to run inference using the model cannot be beyond what the business case justifies.

Trade-off; always and everywhere!

- There is nearly always a trade-off between **cost** and **performance** in terms of **latency and/or throughput**.
- As models become more complex and more and more features are included, the resource requirements increase for every part of the training and serving infrastructure.
- The key is to find the right balance. **Finding the right balance between cost and complexity is a skill that seasoned practitioners build over time.**

Provide what is needed to continue sth

- The prediction request to our ML model might not provide all the features required for prediction. Some of the features might be precomputed or aggregated, and read in real time from a datastore.
- Take the example of a food delivery app that should predict the estimated time for order delivery. This is based on a number of features such as the **list of incoming orders and the number of outstanding orders per minute in the past hour**. Features such as these will be read from a datastore. We will need powerful caches to retrieve this data with low latency, since delivery time has to be updated in real time. We cannot wait for seconds to retrieve data from a database. So, of course, this has cost implications.
- **NoSQL databases** are a good solution to implement caching and feature lookup.

Model Deployments

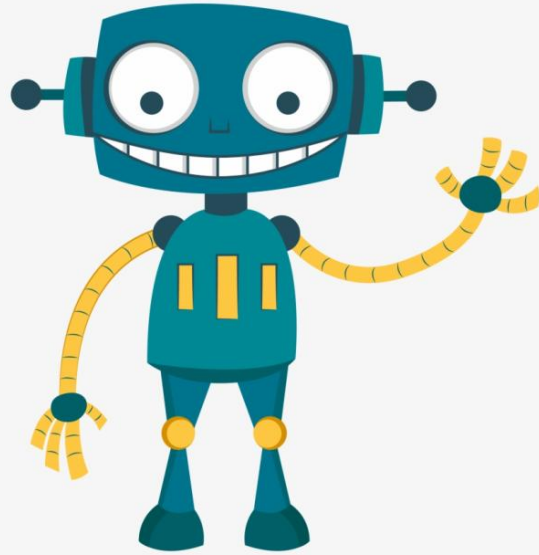
1. We can have a centralized model in a data center that is accessed by a remote call.
2. We can distribute instances of our model to devices that are closer to the end user, such as in a mobile, edge, or embedded system deployment.

Model Servers

Users of our model need a way to make requests. Often this is through a web application that makes calls to a server hosting our model. The model is wrapped as an API service in this approach.

Model Servers

- Both Python (Flask and Django) and Java (Apache Tomcat and Spring) have many web frameworks that can help us achieve this.
- Model servers such as **TensorFlow Serving** can manage model deployment: for example, creating the server instance and managing it to serve prediction requests from clients.
- TensorFlow Serving also offers both the REST and gRPC protocols (gRPC is often more efficient than REST). It can handle up to **100,000 requests per second, per core**, making it a very powerful tool for serving ML applications.



Chapter 12

Model Serving Patterns

How we serves our ML Models?

The primary ways to serve a model are:

1. batch process
2. real-time process.

Batch Inference

- پس از اینکه یک مدل یادگیری ماشین را آموزش داده، ارزیابی کردید و تنظیمش نمودین (تیون کردن)، مدل به محیط عملیاتی (production) منتقل می‌شود تا پیش‌بینی‌ها را تولید کند. در کاربردهایی که تأخیر قابل قبول است، می‌توان از مدل برای ارائه‌ی پیش‌بینی‌ها به صورت دسته‌ای (batch) استفاده کرد؛ پیش‌بینی‌هایی که بعداً در یک مقطع زمانی مشخص روی یک مورد کاربردی اعمال می‌شوند.
- پیش‌بینی مبتنی بر استنتاج دسته‌ای زمانی است که مدل به صورت آفلاین و در قالب یک کار دسته‌ای اجرا می‌شود، معمولن برای تعداد زیادی از نقاط داده، پیش‌بینی‌ها لازم نیست به صورت بلادرنگ تولید شوند.
- کارهای دسته‌ای برای پیش‌بینی معمولن بر اساس یک برنامه‌ی زمان‌بندی تکرارشونده، مثلن روزانه یا هفتگی، اجرا می‌شوند. پیش‌بینی‌ها معمولن در یک پایگاه داده ذخیره می‌شوند و می‌توانند در اختیار توسعه‌دهندگان یا کاربران نهایی قرار بگیرند.
- هنگام انجام پیش‌بینی‌های دسته‌ای، مهم‌ترین معیارهایی که باید بهینه شوند معمولن هزینه و توان عملیاتی (throughput) هستند. **در پیش‌بینی‌های دسته‌ای همیشه باید هدفمان افزایش توان عملیاتی باشد، نه کاهش تأخیر برای پیش‌بینی‌ها بصورت تک به تک.**

Real-Time Inference

- برخلاف پیش‌بینی دسته‌ای، در پیش‌بینی بلادرنگ (real-time) اغلب به زمینه‌ی فعلی مشتری یا هر سیستم و برنامه‌ای که درخواست را ارسال می‌کند، همراه با اطلاعات تاریخی نیاز داریم تا بتوانیم پیش‌بینی را انجام دهیم (context is needed). این کار معمولن مستلزم آن است که داده‌های ورودی با داده‌های تاریخی ترکیب شوند تا درخواست نهایی شکل بگیرد.
- تعداد درخواست‌ها یا کوئری‌ها در هر ثانیه می‌تواند بسته به زمان روز یا روز در هفته بسیار متفاوت باشد. بنابراین منابع ما باید بتوانند در زمان اوج تقاضا افزایش یابند و در صورت امکان کاهش پیدا کنند تا در هزینه صرفه‌جویی شود.
- استنتاج بلادرنگ اغلب یک ضرورت کسب‌وکاری است، چون به شما امکان می‌دهد بر اساس پیش‌بینی‌هایی که با داده‌های جدید انجام می‌شوند، به‌صورت بلادرنگ به اقدامات کاربر پاسخ دهید. این موضوع برای شخصی‌سازی محصولات و خدمات بر اساس درخواست‌های کاربر بسیار مفید است.
- هنگام بررسی گزینه‌هایتان، این نکته را در نظر داشته باشید که به‌عنوان یک قاعده کلی، **تأخیر کمتر برابر با هزینه بیشتر است**. ارائه پیش‌بینی‌های بلادرنگ می‌تواند به‌صورت همگام (synchronous) یا ناهمگام (asynchronous) انجام شود.

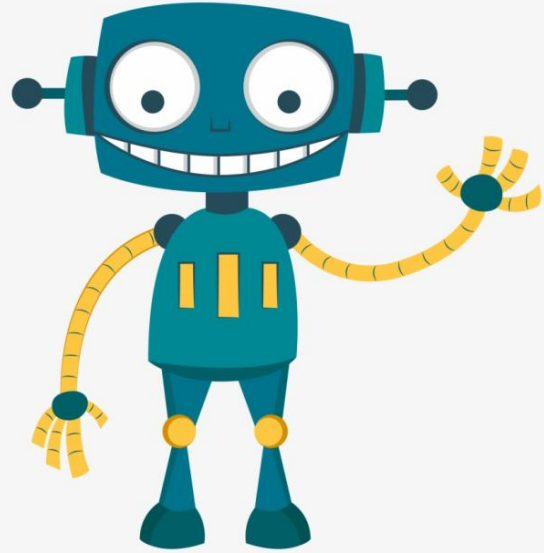
Synchronous Delivery of Real-Time Predictions

کلاينت با يك دروازه يادگيري ماشين (ML gateway) تعامل مي‌کند. اين دروازه به‌عنوان يك مرکز تبادل عمل مي‌کند تا با مدل مستقرشده ارتباط برقرار نمايد، درخواست‌ها را بفرستد و پيش‌بيني‌ها را دريافت کند. درخواست پيش‌بيني و پاسخ (خود پيش‌بيني) **به‌صورت متوالي** بين فراخوان و سرويس مدل يادگيري ماشين انجام مي‌شوند.

Asynchronous Delivery of Real-Time Predictions

پيش‌بيني‌هاي ناهمگام مستقل از درخواست پيش‌بيني به مصرف‌کننده تحويل داده مي‌شوند.

- ارسال فعال (Push): مدل پيش‌بيني‌ها را توليد مي‌کند و آن‌ها را به‌صورت يك اعلان (notification) براي مصرف‌کننده ارسال مي‌کند (مثالي از اين مورد، تشخيص قلب است).
- نمونه‌برداري دوره‌اي (Poll): مدل پيش‌بيني‌ها را توليد مي‌کند و آن‌ها را در يك پايگاه داده ذخيره مي‌کند. مصرف‌کننده به‌صورت دوره‌اي پايگاه داده را براي پيش‌بيني‌هاي تازه‌دردسترس، نمونه‌برداري مي‌کند.



Chapter 15

Model Management and Delivery

MLOps Methodology

Let's look at how MLOps processes evolve and mature as teams become more established and sophisticated.

MLOps Methodology

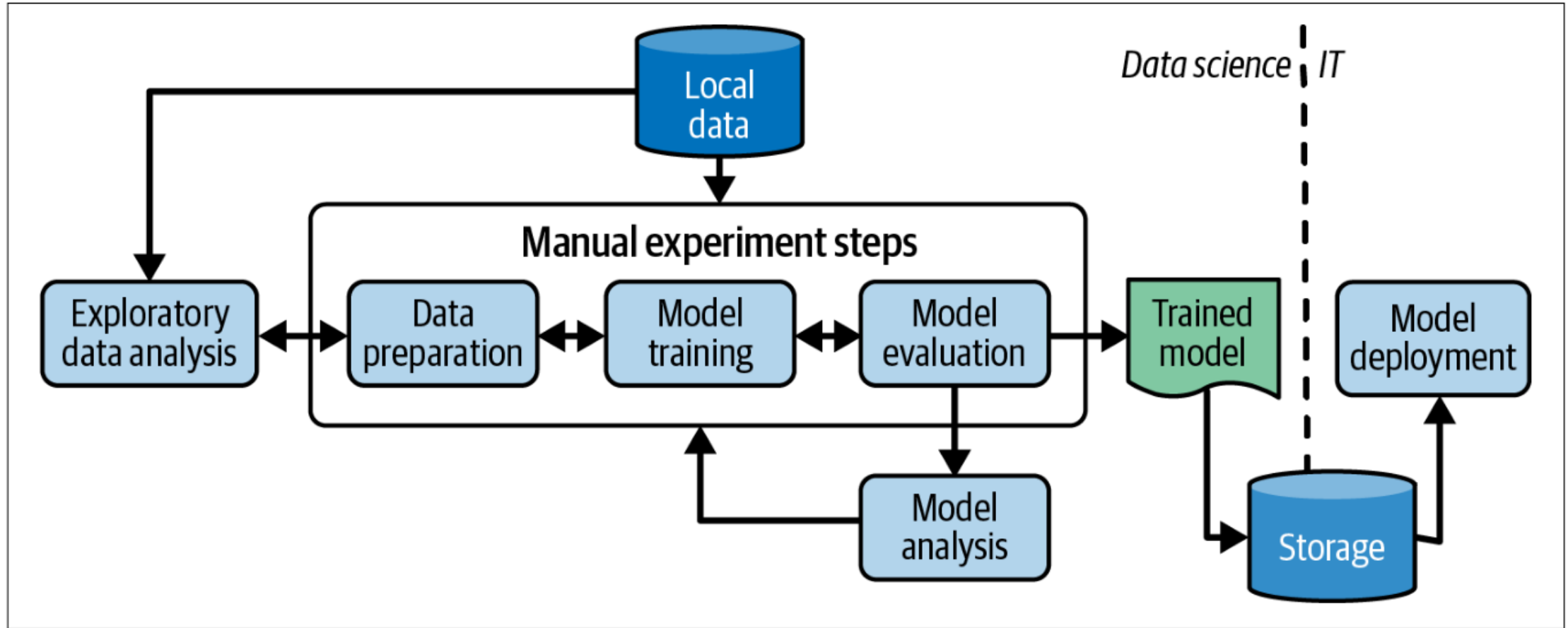
Let's look at how MLOps processes evolve and mature as teams become more established and sophisticated.

MLOps Level 0

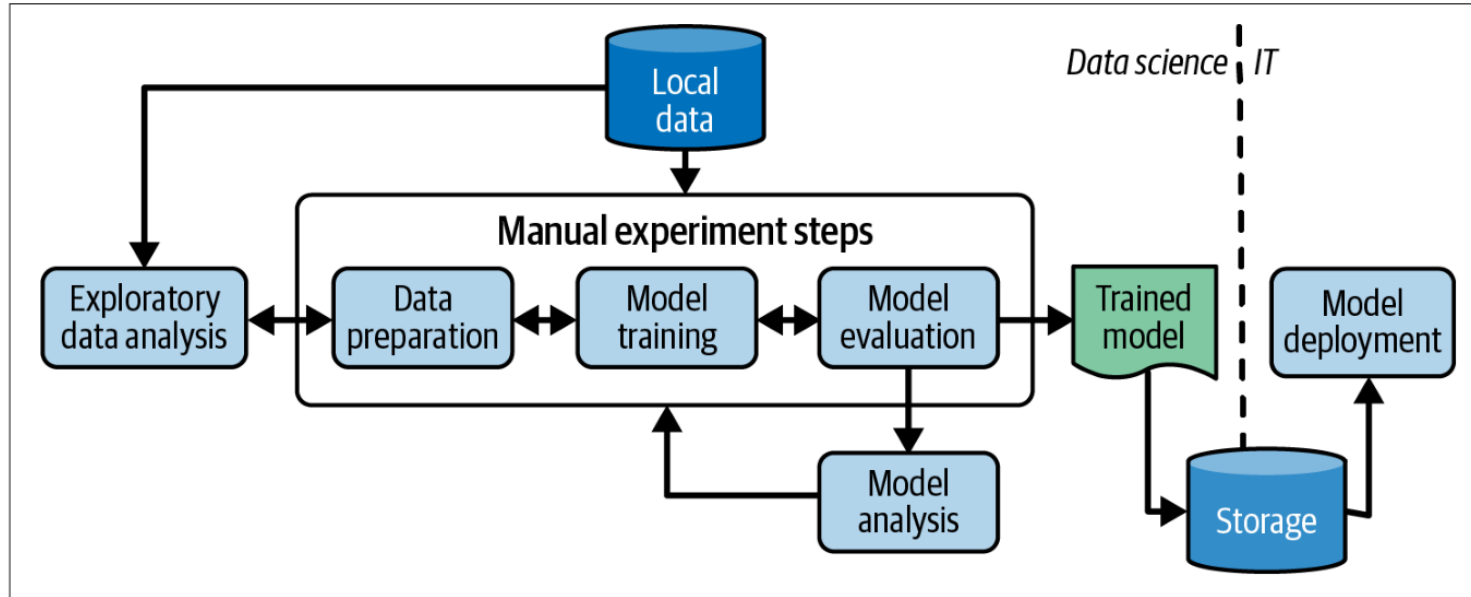
MLOps Level 1

MLOps Level 2

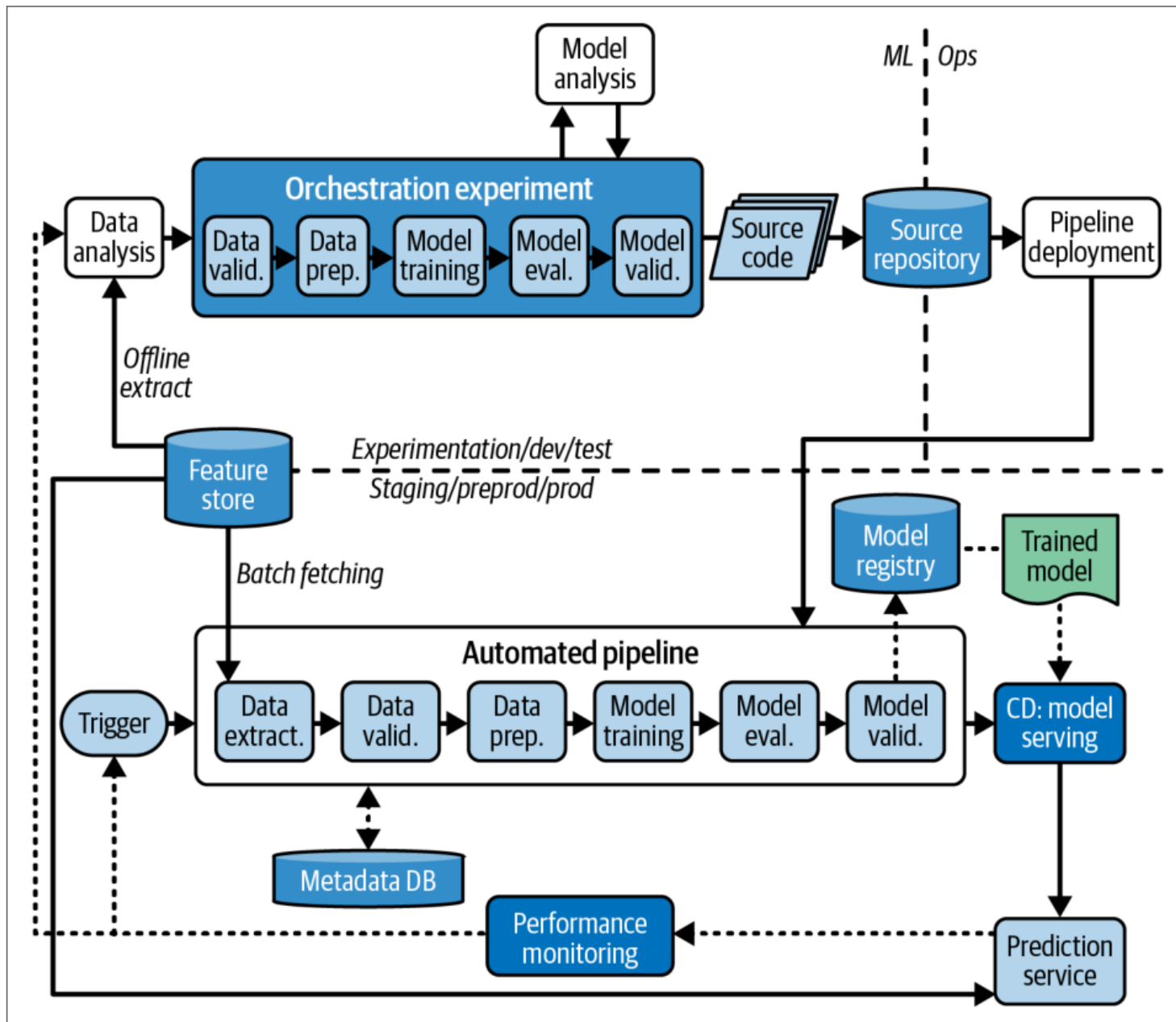
MLOps Level 0



MLOps Level 0



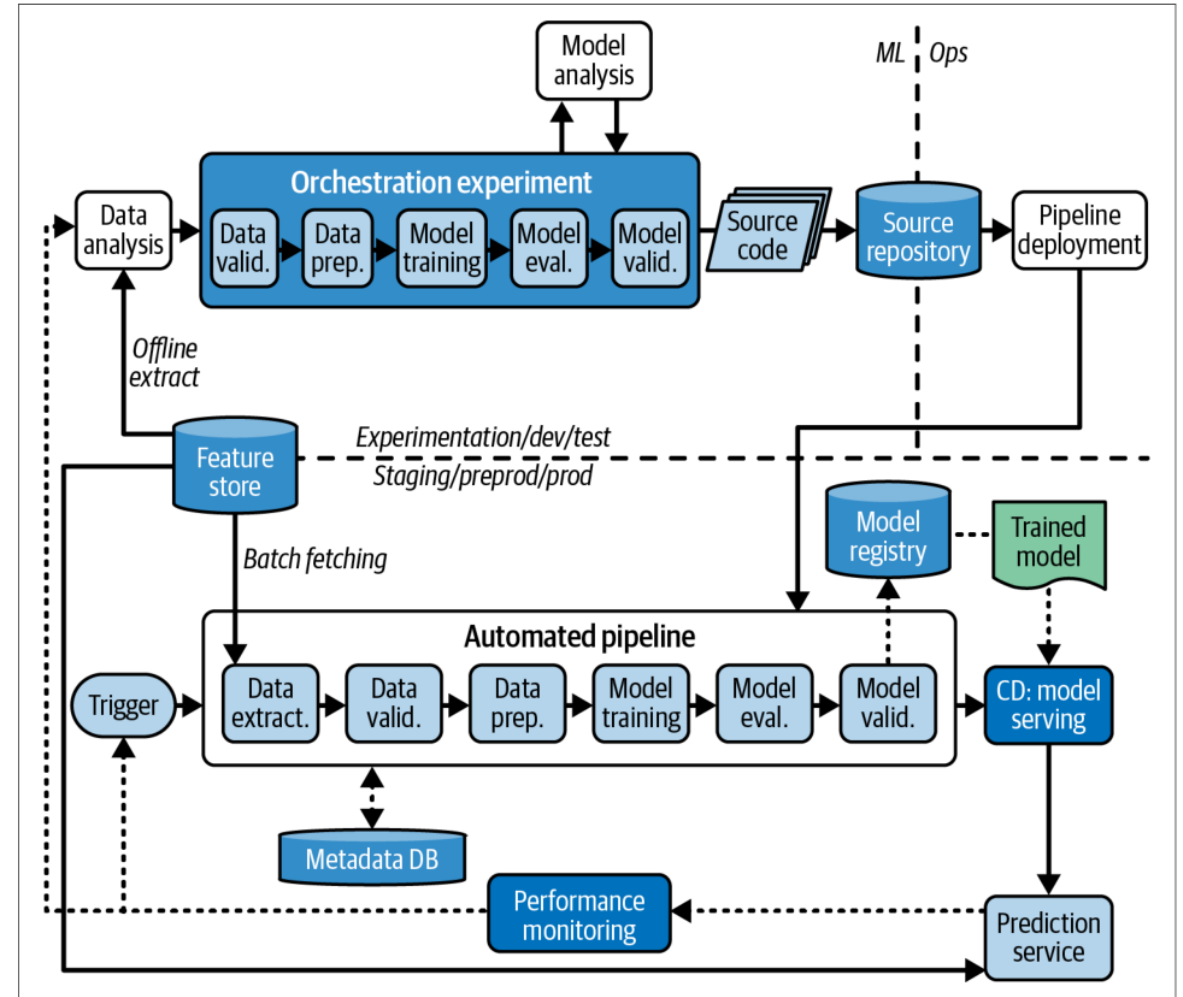
A level 0 process focuses on deploying models, rather than deploying the entire ML system. **It often lacks any monitoring to detect model performance degradation and other model behavioral drifts.** MLOps level 0 is common in many startups and small teams.



MLOps Level 1

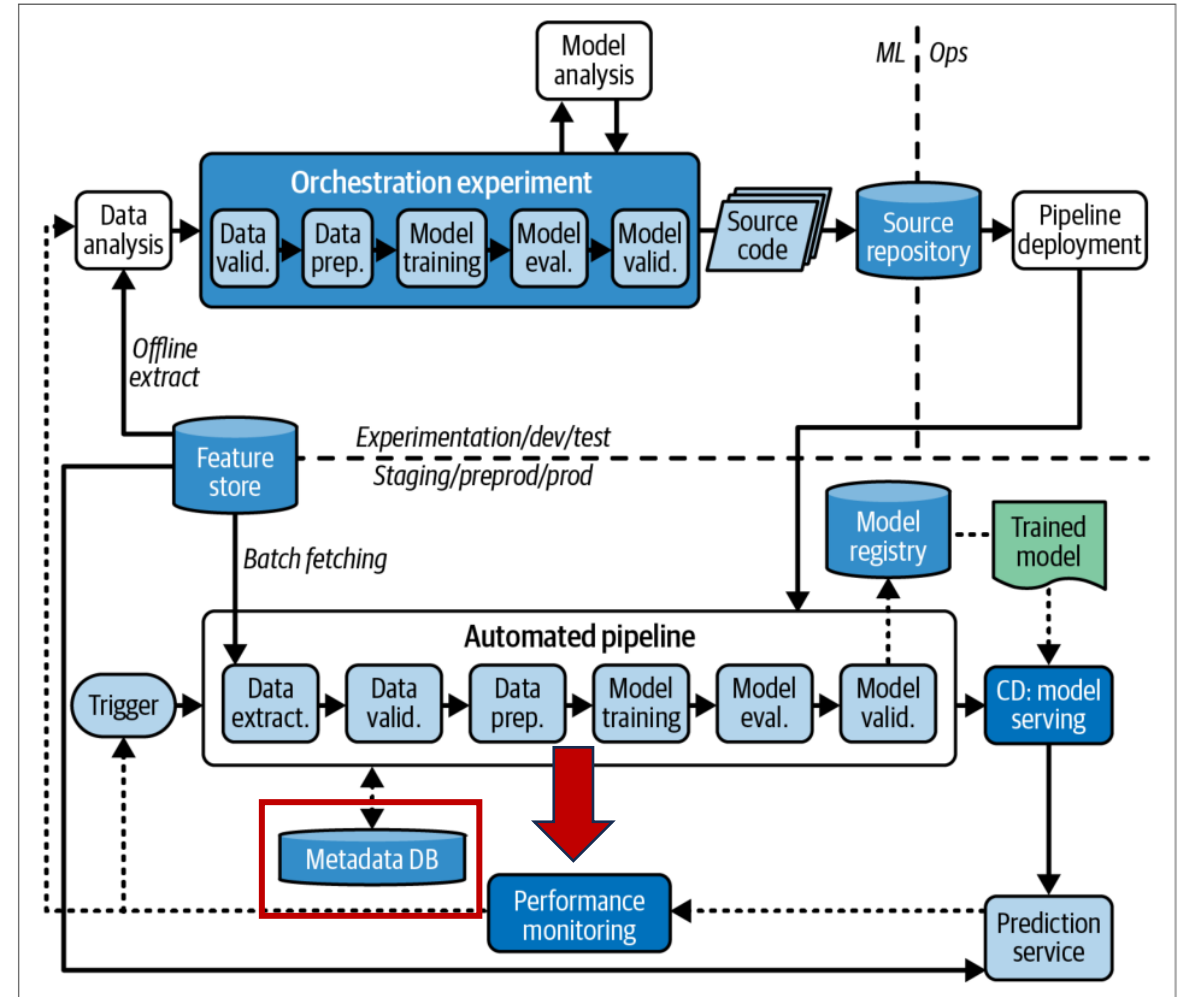
MLOps Level 1

- Automation for continuous training of the model is a primary goal of level 1.
- Another key component of level 1 is the metadata store, where information about each execution of the pipeline is recorded in order to help with data and artifact lineage, reproducibility, and comparisons.

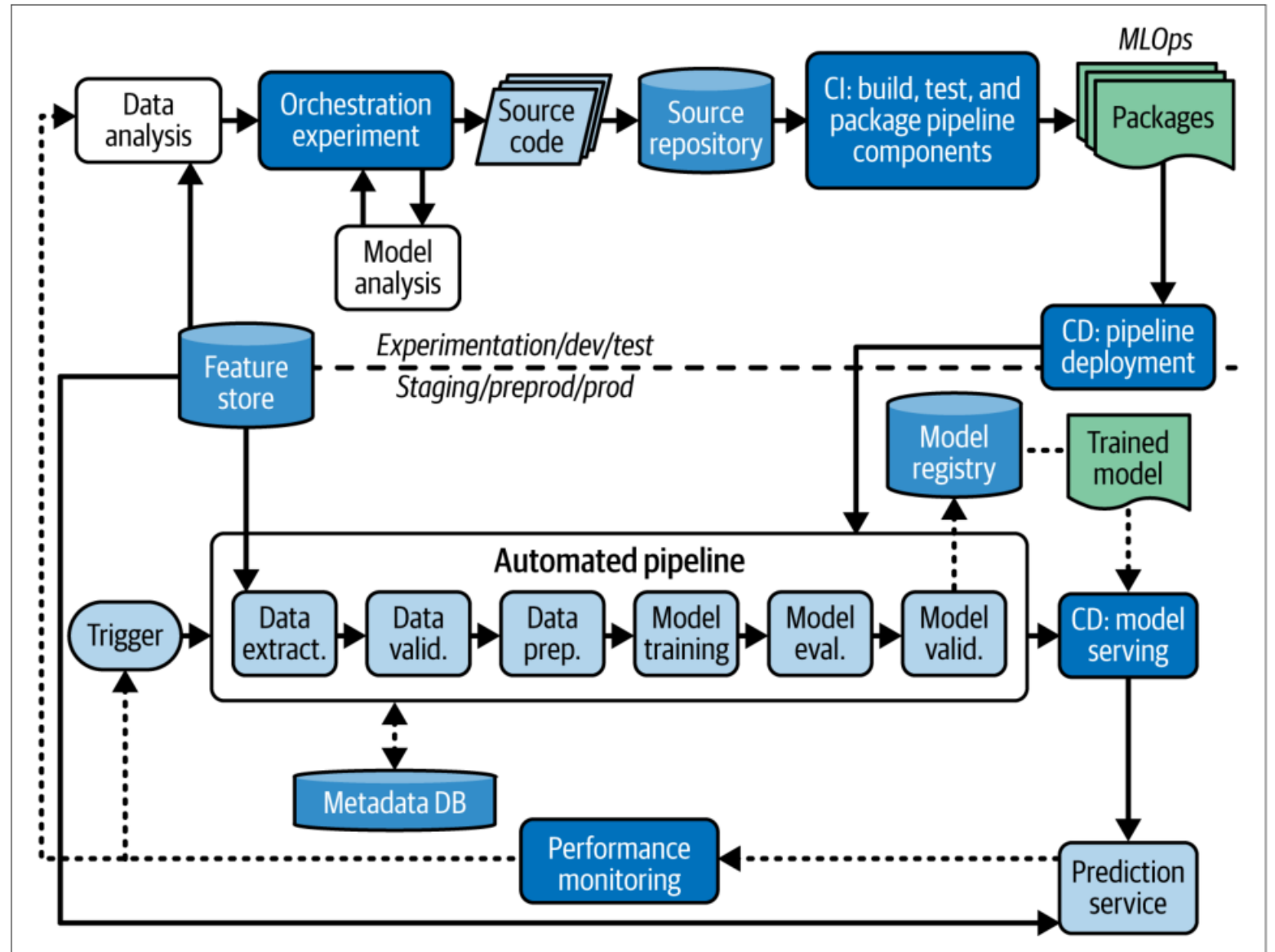


MLOps Level 1

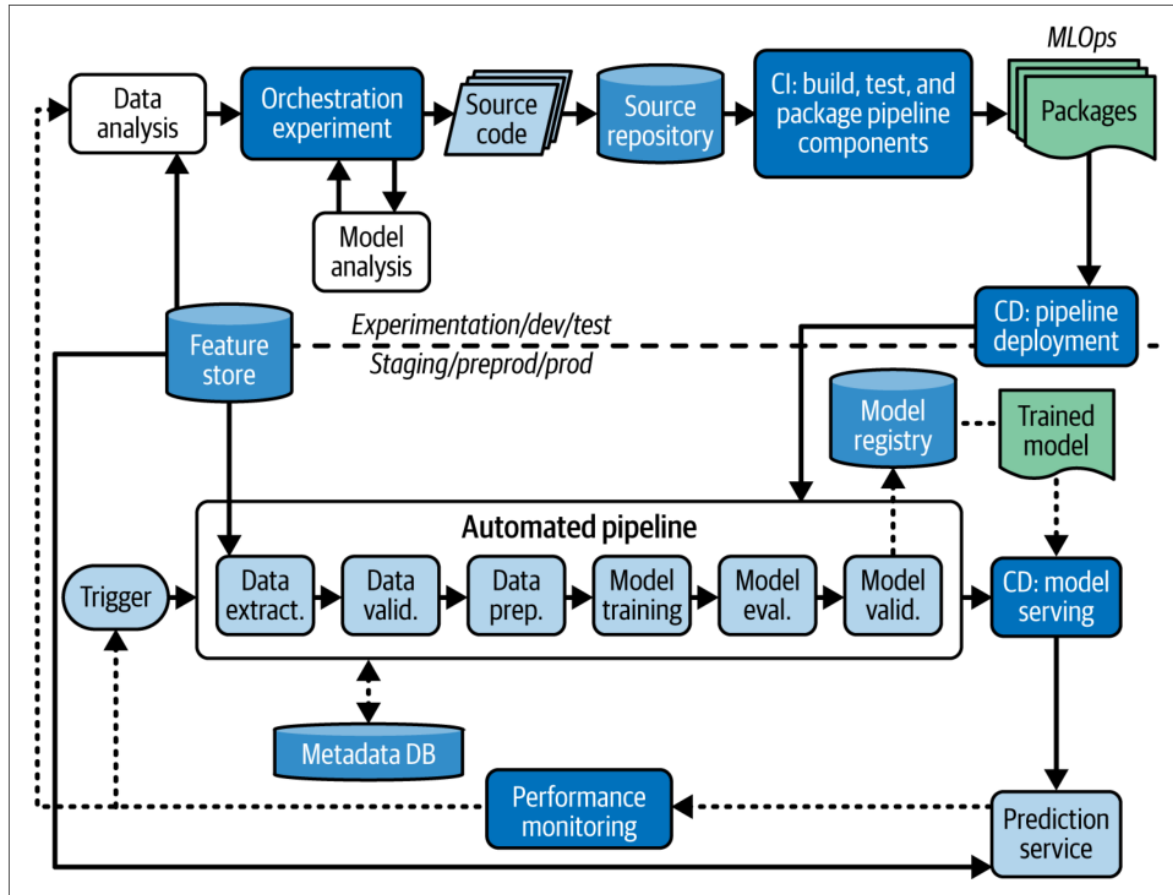
- **Automation** for continuous training of the model is a primary goal of level 1.
- Another key component of level 1 is the **metadata store**, where information about each execution of the pipeline is recorded in order to help with data and artifact lineage, reproducibility, and comparisons.



MLOps Level 2

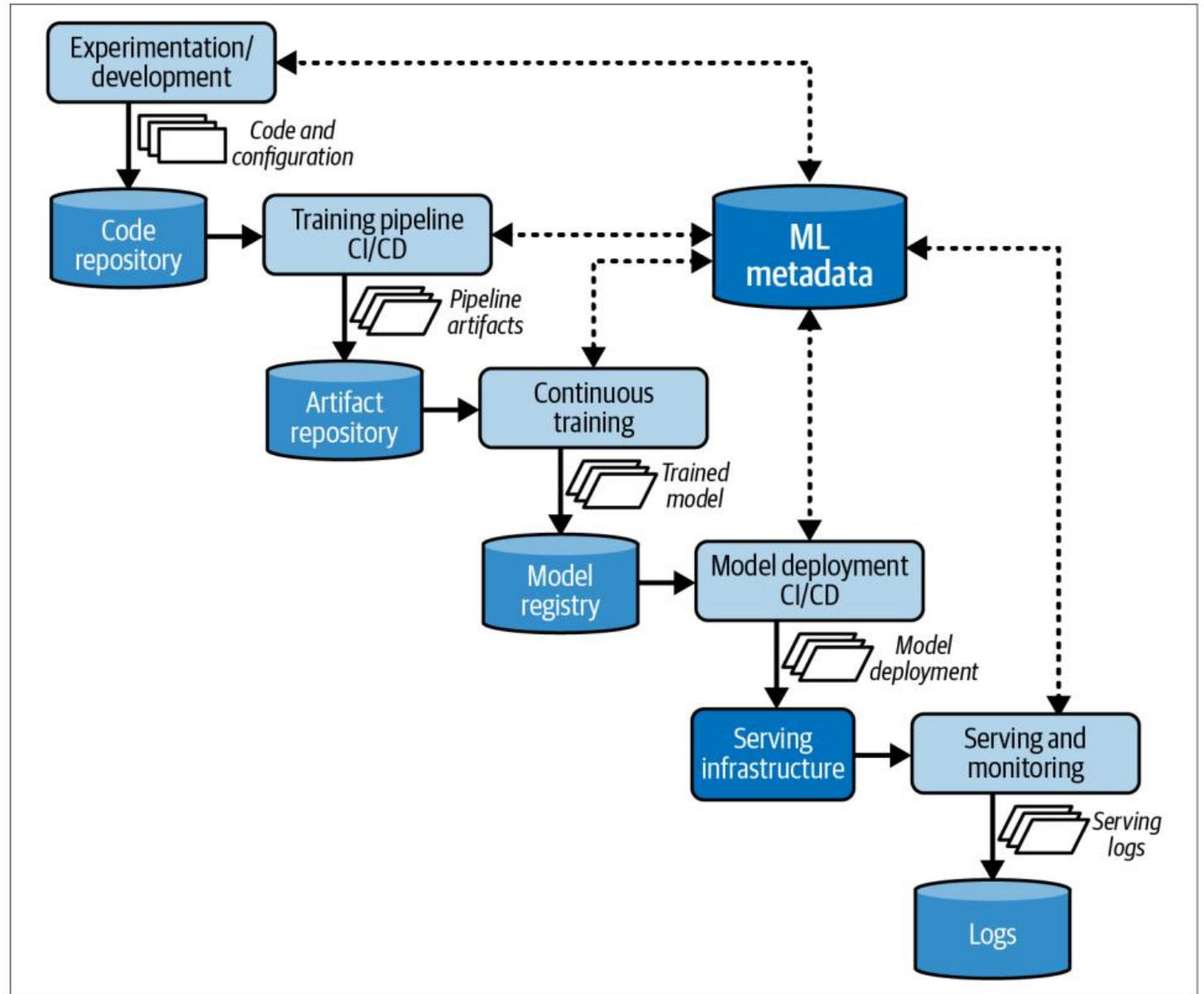


MLOps Level 2



This MLOps setup includes components such as source code control, test and build services, deployment services, a model registry, a feature store, a metadata store, and a pipeline orchestrator.

MLOps Level 2 a simplified view



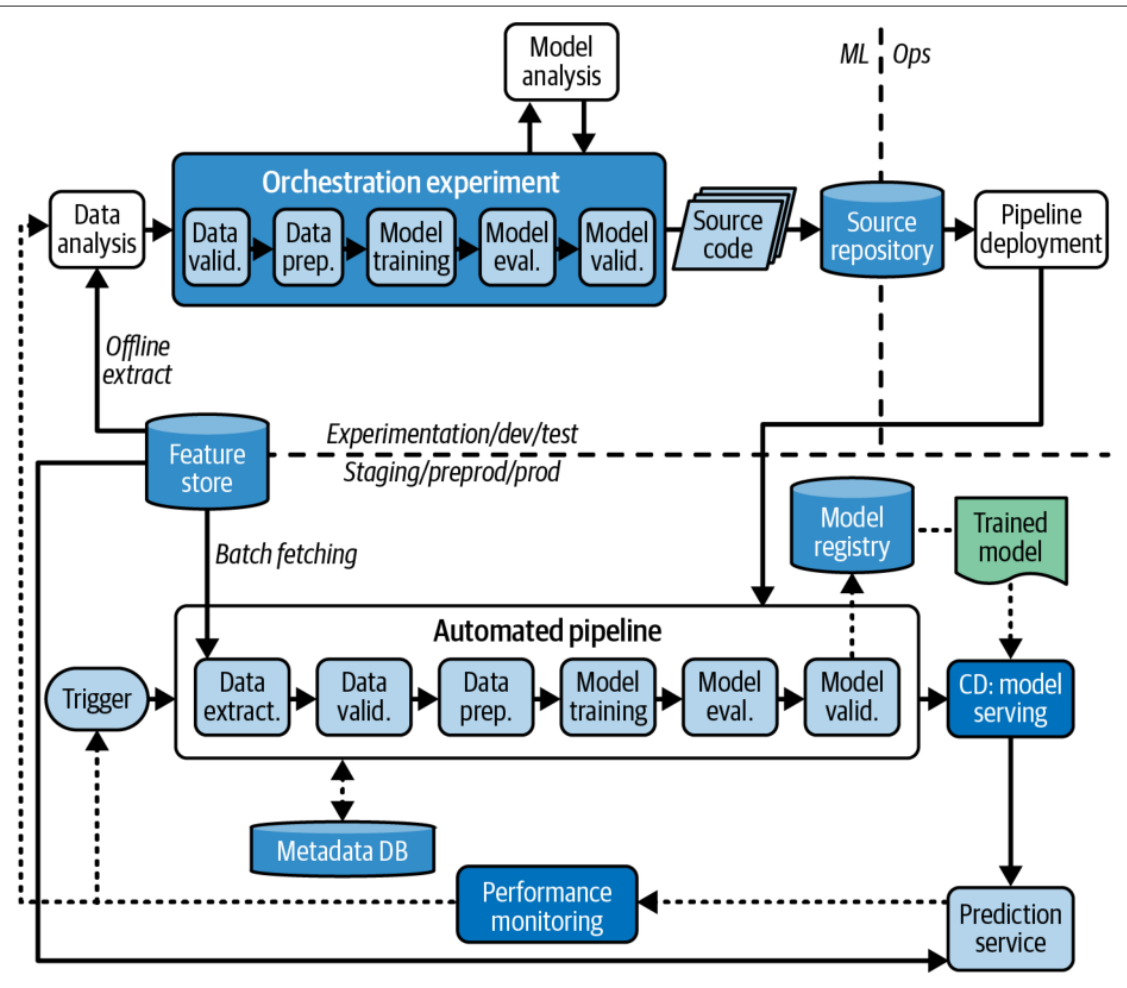
MLOps Different Levels

MLOps Level 0 = Just deploy the model (no monitoring)

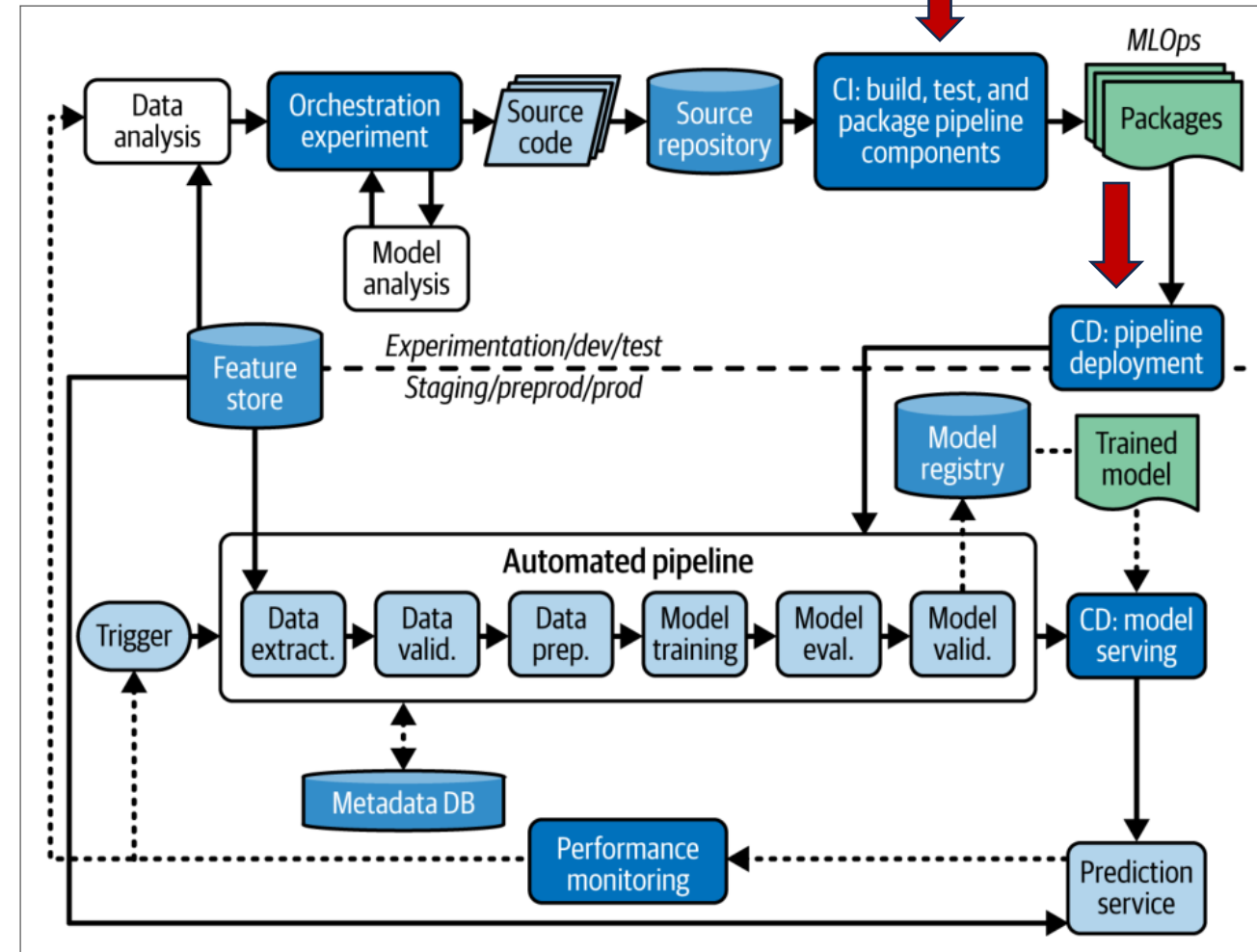
MLOps Level 1 = Automation for Continuous Training

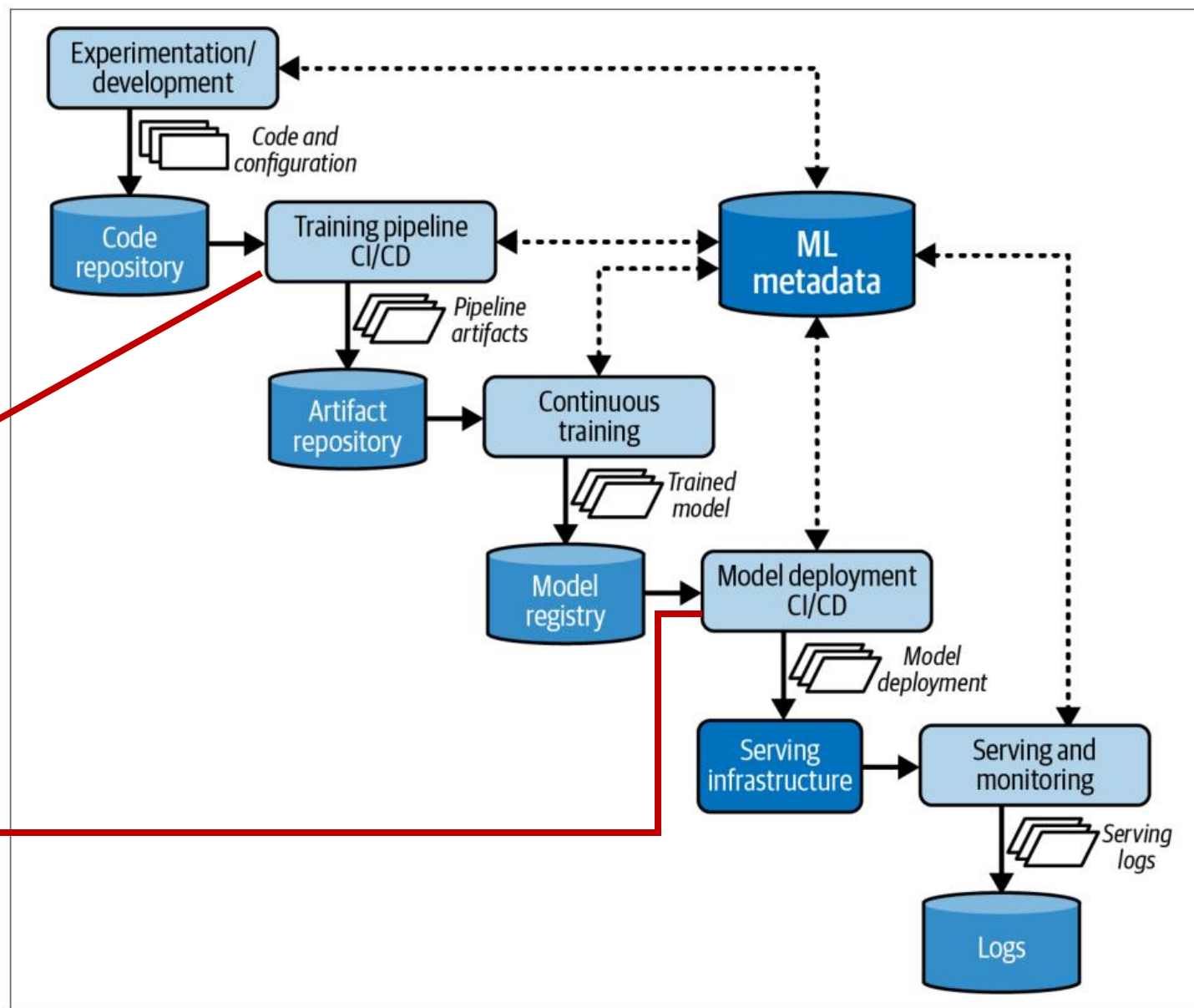
MLOps Level 2 = Automate the whole ML system (CI/CD)

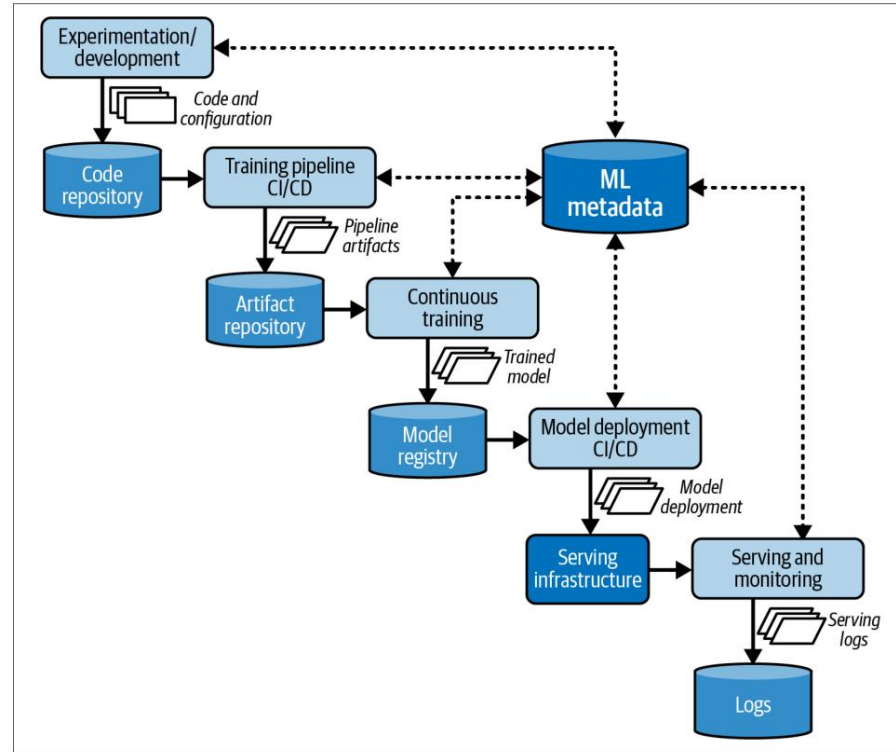
MLOps Level 1



MLOps Level 2



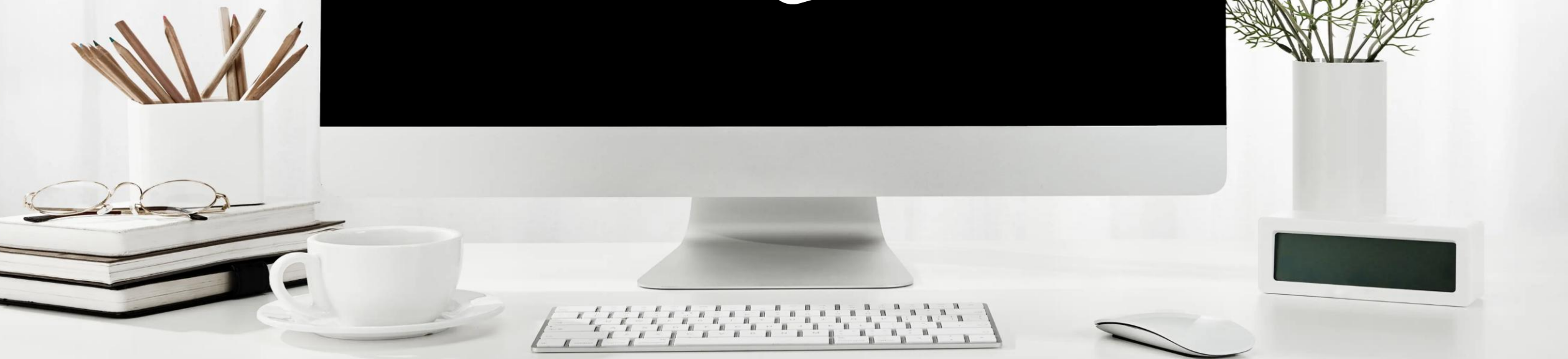


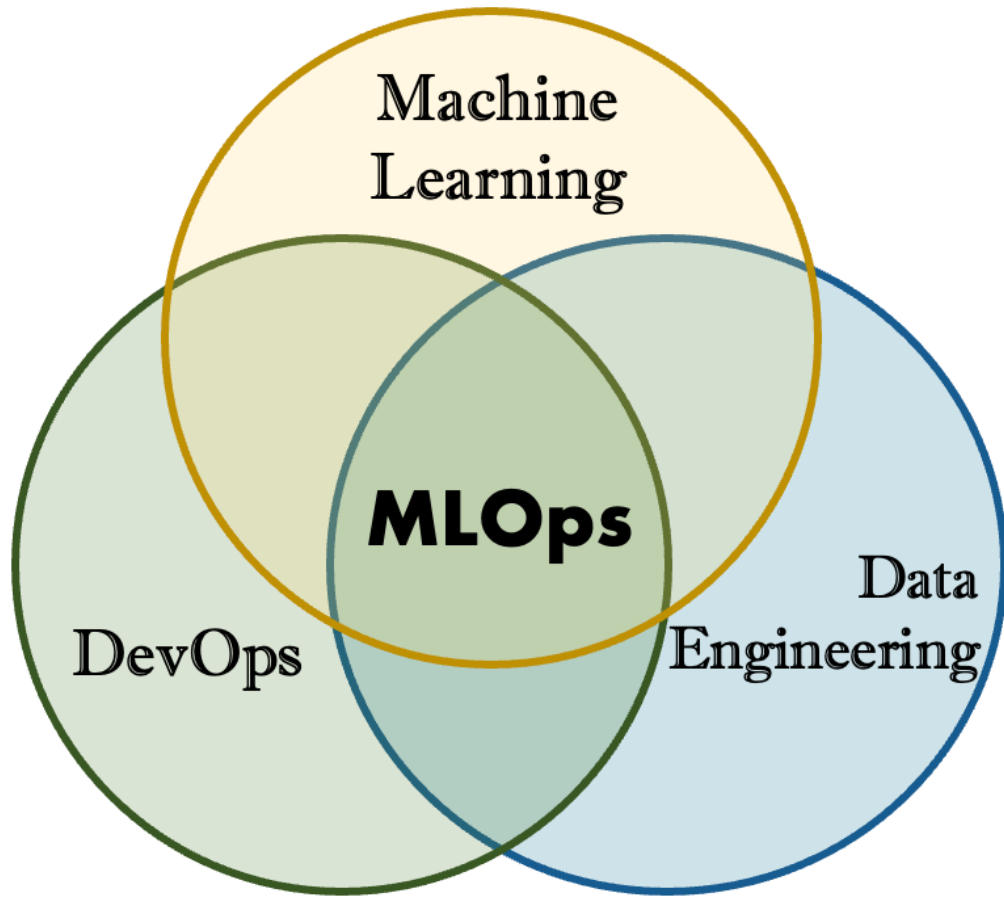


Continuous Deployment: means everything goes automatically.

Continuous Delivery: means the last step of the process will need the approval of one human agent.

پاسخ به سؤال مطرح شده





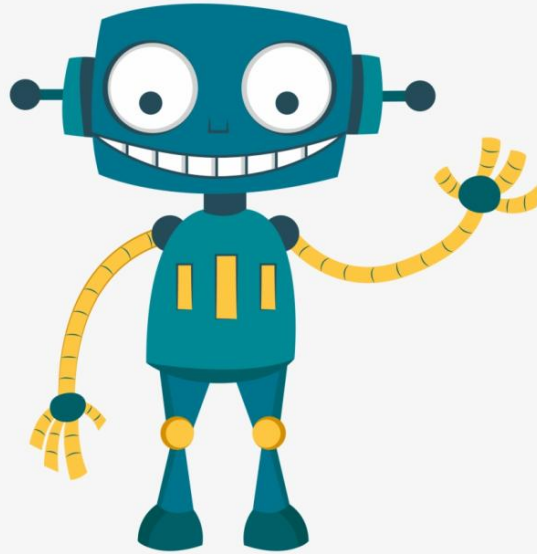
Which one is more **stressful** in production and why?

1- Data Engineering

2- Machine Learning

3- DevOps

Before training our model with sufficient data, business owners take our models to production. And everything FAIL immediately 🤯.



See you all in the next session 🕶️